
A Separation Principle for Cooperative Multi-Agent Reinforcement Learning

Lucia Pezzetti
ETH Zürich
lucia.pezzetti@ai.ethz.ch

Nicolas Lanzetti
Caltech
lnicolas@caltech.edu

Antonio Terpin
ETH Zürich
aterpin@ethz.ch

Florian Dörfler
ETH Zürich
dorfler@ethz.ch

Giorgia Ramponi
University of Zürich
ramponi@ifi.uzh.ch

Abstract

We study cooperative multi-agent reinforcement learning (MARL) problems in which agents evolve under decoupled noisy dynamics but are coupled via a population objective. A canonical example is a mobility operator that orchestrates a fleet of vehicles in an urban environment to meet customer demand. The stochastic and time-varying nature of these problems calls for learning approaches, yet existing MARL algorithms do not scale: learning assignment and routing jointly over the full state-action space becomes prohibitive as the number of agents grows. By studying the problem as a control problem over probability measures, we prove a separation principle: the population-level cost-to-go function is upper-bounded by an optimal transport problem whose transportation cost is the target-conditioned single-agent cost-to-go function. This result simplifies the multi-agent problem to learning a single-agent policy and coordinating the agents via optimal transport. Thus, we introduce SALT (Separation-based Assignment and Learning via optimal Transport) a family of algorithms that learn a target-conditioned single-agent policy using any standard RL algorithm, and then coordinate the agents via optimal transport. Because learning occurs at the single-agent level, SALT scales to fleet sizes where MARL baselines become intractable and transfers without retraining to fleet sizes unseen during training. In a vehicle-delivery experiment in south Manhattan’s road network, where MARL baselines are computationally prohibitive, SALT improves travel times by up to 20% over a shortest-path routing heuristic and transfers without retraining to different fleet sizes. In stochastic grid worlds, SALT reduces average travel times by more than 40% relative to MARL baselines and scales to population sizes where these methods fail.

1 Introduction

Consider a fleet of autonomous vehicles serving delivery requests in a city. At each decision time, the operator must simultaneously decide which vehicle should serve which request and how each vehicle should navigate to its assigned destination. These decisions are, in principle, tightly coupled. Assignments based only on nominal shortest paths can be poor when travel times depend on traffic lights, congestion, and random delays. The best assignment depends on how long it will actually take each vehicle to reach each target, which depends on the routing, and hence on the assignment itself. Yet, solving the full joint assignment-and-routing problem over the fleet quickly becomes computationally prohibitive. Beyond delivery fleets, the same structure arises across a wide range of cooperative systems: inspection drones achieving coverage patterns [1], warehouse robots repositioning to incoming orders [2], emergency response units relocating to optimal positions [3],

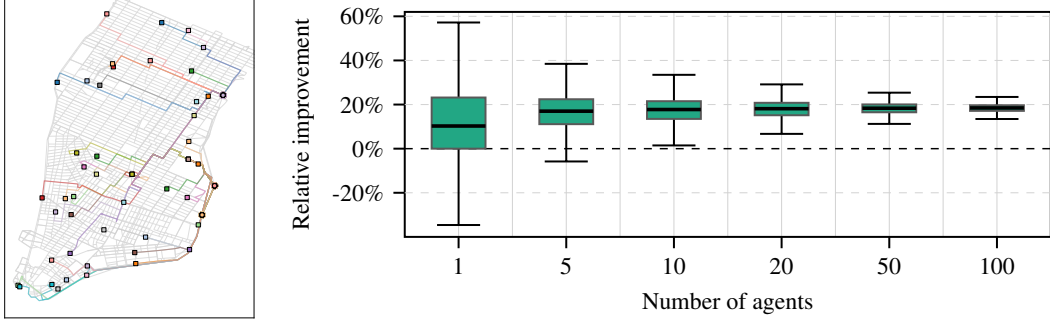


Figure 1: Relative improvement in travel time of SALT over the shortest-path baseline on the time-varying Manhattan road network, as a function of fleet size. Boxplots show the distribution across 500 evaluation instances and 3 seeds, and the dashed line indicates zero improvement. All models are trained with 20 vehicles and evaluated without retraining on fleets of 1-100 vehicles.

ride-hailing platforms balancing supply and demand [4], and electric vehicle sharing charging infrastructure [5]. In all these settings, the goal is to optimally coordinate a fleet of decoupled agents so that its collective terminal state matches or closely approximates a desired target configuration: a spatially uniform coverage, a set of delivery locations, or a demand-weighted positioning pattern. These problems are typically solved in a receding-horizon fashion to account for changing conditions and request patterns and, thus, they require modular approaches that generalize without retraining when the fleet size or target distribution change.

A standard approach for these problems is multi-agent reinforcement learning (MARL) [6, 7, 8, 9], often under centralized training with decentralized execution. Representative approaches include policy-gradient methods such as MAPPO [10] and MADDPG [11] and value-decomposition methods such as VDN [12], QMIX [13], and QPLEX [14]. Although effective in many settings [15, 16], their learning complexity therefore grows prohibitively with the number of agents [17], limiting the scalability and generalization to fleet sizes not seen during training. Value-decomposition methods gain tractability by imposing architectural assumptions on the joint value function, such as additivity or monotonicity, which however hold only in very specific settings [18]. Mean-field reinforcement learning [19, 20] reduces complexity by replacing pairwise interactions with an aggregate representation of the population. Despite being practical in large systems [4, 21], mean-field approaches rely on approximations that can lose fidelity when modeling finite populations [22]. A separate line of work addresses assignment via combinatorial solvers or attention-based policies [23, 24], but treats routing as a downstream heuristic step. Such methods fail to take into account that a correct assignment cost is itself a stochastic control quantity. Heuristics can therefore assign vehicles to targets that “look close” but are expensive in expectation.

In this work, we seek a principled way to separate the population-level assignment and the single-agent control problem. Inspired by the “deterministic” separation principle for multi-agent control in [25], we cast the optimal control problem for the population as an optimal control problem over the space of probability measures where the control costs are optimal transport costs (e.g., Wasserstein distance) to a desired target measure. By construction, this representation is invariant to the permutations and can capture both finite populations (as empirical measures) and a continuum of agents (as continuous measures). In the deterministic setting, [25] shows that the dynamic programming recursion over the space of probability measures can be separated into an optimal transport problem and a single-agent control problem. However, this result does not extend exactly to the stochastic setting. Nevertheless, we show here that a carefully designed separation principle, where noise is “averaged out”, continues to upper bound the control cost. This analysis is non-trivial: In the stochastic setting, the probability measure capturing the population is itself random, yielding a probability measures over probability measures—a modeling challenge itself, compounded by delicate measurability issues. Finally, differently from [25], we do not limit ourselves to a theoretical analysis, but leverage our separation principle to design scalable algorithms for MARL.

Contributions. Our contributions are as follows:

1. We formulate cooperative MARL with homogeneous agents, decoupled stochastic dynamics, and terminal optimal transport formation costs as an optimal control problem over probability

measures. This formulation removes artificial agent labels, applies to both finite populations and continuum of agents, is naturally invariant to permutations of the agents, and captures terminal formation or coverage costs through optimal transport.

2. We prove the separation principle for this optimal control problem (Theorem 3.1, Part I). The population-level cost-to-go function is upper-bounded by an optimal transport problem whose transportation cost is the target-conditioned single-agent cost-to-go function. The resulting problem decouples the allocation problem (the transportation costs are fixed) from the routing problem (which can be solved independently from the fleet). In the deterministic setting, this upper bound becomes exact, recovering the deterministic separation principle from [25, Corollary 4.2]. Importantly, our proof is constructive and directly yields an implementable policy.
3. This separation yields SALT, a family of MARL algorithms that learn a shared target-conditioned single-agent policy and coordinate agents via optimal transport. SALT is agnostic to the choice of the single-agent algorithm and generalizes to population sizes not seen during training.
4. We validate SALT on a time-varying south Manhattan road network with 2088 intersections and 4262 directed roads. SALT trained with $N = 20$ vehicles transfers without retraining to fleets of 1-100 vehicles and reduces travel time by up to 20% relative to a nominal shortest-path routing heuristic. We also use stochastic grid worlds for controlled ablations. SALT variants outperform standard MARL baselines by more than 40% in average travel time, consistently across different single-agent RL algorithms. Additionally, SALT greatly improves sample efficiency: with single-agent PPO, it needs up to $450\times$ fewer environment interactions than MAPPO and scales to $N = 128$, while MAPPO already fails at $N = 8$ within the 2×10^9 -interaction budget.

Notation. We work with Polish spaces, Borel functions, and Borel probability measures. For a Polish space X , $\mathcal{P}(X)$ denotes the space of probability measures on X . We write δ_x for the Dirac measure at $x \in X$. The projection from $X \times Y$ onto X is denoted as $\text{proj}_X^{X \times Y}$. For $\mu \in \mathcal{P}(X)$ and $T: X \rightarrow Y$, the pushforward of μ through T , $T_\# \mu \in \mathcal{P}(Y)$, is defined by $T_\# \mu(B) = \mu(T^{-1}(B))$ for all Borel sets $B \subseteq Y$. A coupling between the probability measures $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is a joint probability measure with marginals μ and ν ; the set of couplings between μ and ν is $\Gamma(\mu, \nu)$. For a nonnegative cost $c: X \times Y \rightarrow \mathbb{R}_{\geq 0}$, the optimal transport cost between μ and ν is

$$\mathcal{K}[c](\mu, \nu) := \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times Y} c(x, y) d\gamma(x, y).$$

If $k: X \rightarrow \mathcal{P}(Y)$ is a Markov kernel, $\mu \otimes k$ denotes the induced joint law on $X \times Y$. In particular, if $\pi(dx, dy) = \mu(dx)k(x; dy)$ is a disintegration, we write $\pi = \mu \otimes k^x$.

2 Problem setting

We consider a class of cooperative MARL problems in which N homogeneous agents coordinate to reach a desired configuration over a finite horizon T , while incurring agent- and terminal population-level costs. Let S , A , W , and Z denote the state, action, noise, and target spaces. While $Z = S$ in the canonical target matching, we keep Z distinct for generality: targets may be lower-dimensional or task-level descriptors, such as destinations when the state also includes velocity or battery level.

2.1 Agent-based formulation

Each agent i has state $s_t^i \in S$ and chooses action $a_t^i \in A$. Its noisy dynamics are

$$s_{t+1}^i = f_t(s_t^i, a_t^i, w_t^i), \quad w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i),$$

where $\xi_t(\cdot | s_t^i, a_t^i) \in \mathcal{P}(W)$ gives the conditional law of the noise affecting an agent in state s taking action a . We consider three noise models, which differ in how noise is correlated across agents: *individual noise*; where noise is sampled independently across agents, *local noise*; where agents with the same state-action pair share the same noise realization, and *global noise*, where all agents share a common noise realization. Their formal definition can be found in Appendix A. Let $\mathbf{s}_t := (s_t^1, \dots, s_t^N)$ and $\mathbf{a}_t := (a_t^1, \dots, a_t^N)$ denote the ordered agents' states and actions. The control cost results from the sum of agent-level stage costs $g_t: S \times A \times W \rightarrow \mathbb{R}_{\geq 0}$ and a population-level terminal formation cost. The terminal cost compares the final population distribution

$\mu_T = \frac{1}{N} \sum_{i=1}^N \delta_{s_T^i}$ to a target configuration $\nu \in \mathcal{P}(Z)$ through the optimal transport cost

$$\mathcal{K}[g_T](\mu_T, \nu) = \inf_{\gamma \in \Gamma(\mu_T, \nu)} \int_{S \times Z} g_T(s, z) d\gamma(s, z), \quad (1)$$

where $g_T : S \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$ is the transportation cost; e.g., when $g_T(s, z) = \|s - z\|^p$ for $p \geq 1$, then $\mathcal{K}[g_T](\mu_T, \nu)$ is the p power of the type- p Wasserstein distance. Since the problem is a fully observed cooperative MARL problem, it can be cast as a Multiagent MDPs [26], which reduces to a standard single-agent MDP over the joint action space. By standard MDP theory [27], we losslessly consider only deterministic (measurable) Markov policies $\pi = (\pi_t)_{t=0}^{T-1}$. We denote this class by Π .

Problem 1. Given initial states $\bar{s}_0 = (\bar{s}_0^1, \dots, \bar{s}_0^N)$ and a target distribution ν , learn a policy $\pi \in \Pi$ minimizing the expected cumulative cost

$$J^N(\bar{s}_0, \nu) = \inf_{\pi \in \Pi} \mathbb{E}_{(w_t)_{t=0}^{T-1}} \left[\frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T-1} g_t(s_t^i, a_t^i, w_t^i) + \mathcal{K}[g_T](\mu_T, \nu) \right]$$

s.t. $s_{t+1}^i = f_t(s_t^i, a_t^i, w_t^i), \quad s_0 = \bar{s}_0.$

2.2 Population-based formulation

We generalize Problem 1 at the *population-level* by replacing the ordered state s_t with the empirical measure $\mu_t = \frac{1}{N} \sum_{i=1}^N \delta_{s_t^i} \in \mathcal{P}(S)$ which discards agent labels and makes the permutation invariance in the problem explicit. For now, we allow general population states $\mu_t \in \mathcal{P}(S)$; we will later specialize this formulation back to the finite-agent setting.

Population control. A population-level control must specify how the mass at each state is split among actions. Such a control is a *state-action distribution* $\lambda_t \equiv \lambda_t(\mu_t) \in \mathcal{P}(S \times A)$ whose S -marginal is the current population state: $(\text{proj}_S^{S \times A})_{\#} \lambda$, the pushforward via the projection map $\text{proj}_S^{S \times A}$. Let

$$M(\mu_t) := \{\lambda \in \mathcal{P}(S \times A) : (\text{proj}_S^{S \times A})_{\#} \lambda = \mu_t\}$$

denote the set of admissible state-action distributions at population state μ_t . For example, if all agents follow a common deterministic policy $\pi_t : S \rightarrow A$, then $\lambda_t = (\text{Id}_S, \pi_t)_{\#} \mu_t$ that is, each unit of mass at state s is placed at the state-action pair $(s, \pi_t(s))$.

Noise and population dynamics. When dynamics are deterministic, the next population state is the pushforward of μ under the dynamics, i.e., $\mu_{t+1} = f_t_{\#} \lambda_t$. The stochastic case, however, requires more care. The transition also depends on the specific noise realizations across the population. In the agent-based formulation, agent i simply draws $w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i)$ and transitions to $f_t(s_t^i, a_t^i, w_t^i)$. At the population level, however, we cannot simply replace these individual draws by the underlying distribution $\xi_t(\cdot | s, a)$: the next population state is determined by the *realized collection* of noise draws across all agents, which is itself a random object. These draws are random with distribution conditioned over state-action pairs and hence need to be represented as a state-action dependent *random* probability measure $\hat{\xi}_t^{s,a} \in \mathcal{P}(W)$. The distinction between $\xi_t(\cdot | s, a)$ and $\hat{\xi}_t^{s,a}$ is crucial: the former is the single-agent noise law, while the latter collects the noise realizations that drive the population transition. Collecting the realized empirical kernels $\hat{\xi}_t^{s,a}$ across all state-action pairs, we define the joint state-action-noise distribution $\rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a} \in \mathcal{P}(S \times A \times W)$. The distribution ρ_t records exactly the fraction of the population experiencing each triple (s, a, w) . The population then evolves according to the pushforward of the joint state-action-noise distribution ρ_t under the dynamics f_t , so that $\mu_{t+1} = f_t_{\#} \rho_t$.

Example 2.1. Consider three agents with the state-action distribution $\lambda_t = \frac{2}{3} \delta_{(0,a)} + \frac{1}{3} \delta_{(1,a)}$, i.e., two agents have state 0 and use action a , one agent has state 1 and uses action a . Assume that agents are subject to individual noise and that the noise realizations for the two agents at $(0, a)$ are 0 and 1, while the noise realization for the agent at $(1, a)$ is 1. Then, $\hat{\xi}_t^{0,a} = \frac{1}{2} \delta_0 + \frac{1}{2} \delta_1$ and $\hat{\xi}_t^{1,a} = \delta_1$. Accordingly, the state-action-noise distribution is $\rho_t := \lambda_t \otimes \hat{\xi}_t^{s,a} = \frac{1}{3} \delta_{(0,a,0)} + \frac{1}{3} \delta_{(0,a,1)} + \frac{1}{3} \delta_{(1,a,1)}$ and the next population distribution is $\mu_{t+1} = (f_t)_{\#} \rho_t = \frac{1}{3} \delta_{f_t(0,a,0)} + \frac{1}{3} \delta_{f_t(0,a,1)} + \frac{1}{3} \delta_{f_t(1,a,1)}$.

Cost. As $\frac{1}{N} \sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) = \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w)$, the control cost is

$$J(\bar{\mu}_0, \nu) = \mathbb{E}_{(\hat{\xi}_t)_{t=0}^{T-1}} \left[\sum_{t=0}^{T-1} \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + \mathcal{K}[g_T](\mu_T, \nu) \right],$$

where $\hat{\xi}_t = (\hat{\xi}_t^{s,a})_{(s,a) \in S \times A}$ collects all realized noise kernels at time t . The expectation is then taken over the randomness of these profiles. This keeps finite-population fluctuations and noise correlations explicit while reducing the population-level randomness to the single object $\hat{\xi}_t$. We refer the reader to Appendix A for additional details and explicit construction in canonical noise models.

Problem 2. Given an initial population distribution $\bar{\mu}_0$, find a sequence of state-action distributions $(\lambda_t)_{t=0}^{T-1}$ that minimizes the expected cumulative cost

$$\begin{aligned} J(\bar{\mu}_0, \nu) = \inf_{(\lambda_t)_{t=0}^{T-1}} \mathbb{E}_{(\hat{\xi}_t)_{t=0}^{T-1}} & \left[\sum_{t=0}^{T-1} \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + \mathcal{K}[g_T](\mu_T, \nu) \right] \\ \text{s.t. } \mu_0 = \bar{\mu}_0, \quad \mu_{t+1} = f_{t\#} \rho_t, \quad (\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t \quad \rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a}. \end{aligned}$$

The population formulation removes the artificial labels from the state description and generalizes Problem 1. Any finite-agent policy induces an empirical state-action distribution $\lambda_t = \frac{1}{N} \sum_{i=1}^N \delta_{(s_t^i, a_t^i)} \in M(\mu_t)$, so every policy admissible for Problem 1 is feasible in Problem 2 and $J(\bar{\mu}_0, \nu) \leq J^N(\bar{s}_0, \nu)$. The converse inequality does not hold in general. Problem 2 allows for arbitrary population state-action distributions, which may split mass in ways that are not realizable by N agents. However, we show in Corollary 3.3 that the separation bound derived for Problem 2 directly upper bounds the agent-based cost to go $J^N(\bar{s}_0, \nu)$, yielding a deployable policy for Problem 1.

3 Separation principle

In this section, we prove a separation principle for Problem 2, showing that the population cost-to-go function separates into optimal single-agent control toward a target and optimal target assignment across the population; see Theorem 3.1 below. We then specialize the result for the finite-agent value J^N of Problem 1 and obtain a deployable policy that we will later use for algorithm design.

Target-conditioned single-agent problem. Fix a target $z \in Z$. For an agent in state $s \in S$ at time t , define $j_t(s, z)$ as the optimal cost to go to reach z over the remaining horizon:

$$\begin{aligned} j_t(s, z) = \inf_{(\pi_\tau)_{\tau=t}^{T-1}} \mathbb{E}_{(w_\tau)_{\tau=t}^{T-1}} & \left[\sum_{\tau=t}^{T-1} g_\tau(s_\tau, a_\tau, w_\tau) + g_T(s_T, z) \right], \\ \text{s.t. } s_t = s, \quad s_{\tau+1} = f_\tau(s_\tau, a_\tau, w_\tau), \quad w_\tau \sim \xi_\tau(\cdot | s_\tau, a_\tau). \end{aligned} \quad (2)$$

This is a standard single-agent stochastic optimal control problem—whose complexity is independent of the population size N —that can be studied via the dynamic programming principle:

$$j_t(s_t, z) = \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s_t, a_t, w_t) + j_{t+1}(f_t(s_t, a_t, w_t), z)], \quad j_T(s_T, z) = g_T(s_T, z). \quad (3)$$

A separation principle. For $\mu_t \in \mathcal{P}(S)$, let $J_t(\mu, \nu)$ be the optimal expected cost from time t to the horizon, when the population starts from μ_t and aims to match the target distribution ν . At $t = T$, $J_T(\mu, \nu) = \mathcal{K}[g_T](\mu, \nu)$. Fix $t < T$. Since $\mathcal{P}(S)$ is a Polish space whenever S is Polish [28, Theorem 15.15], the cost is additive over time, and the population dynamics are Markovian, standard dynamic programming yields the population-level Bellman recursion [29, Ch. 8]

$$J_t(\mu_t, \nu) = \inf_{\lambda_t \in M(\mu_t)} \mathbb{E}_{\hat{\xi}_t} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + J_{t+1}(f_{t\#} \rho_t, \nu) \right]. \quad (4)$$

We now state the main result of this work.

Theorem 3.1 (Separation principle, Part I: upper bound for the population cost to go). *Consider Problem 2. For all $t \in \{0, \dots, T\}$, the population cost-to-go function (4) is upper bounded by an optimal transport problem whose transportation cost is the single-agent cost-to-go function j_t in (2):*

$$J_t(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma(s, z). \quad (5)$$

In particular, the inequality holds true for the cost to go $J(\bar{\mu}_0, \nu)$ in Problem 2.

Discussion. The upper bound (5) separates the stochastic population control problem into a single-agent control problem yielding a population-level matching problem in which j_t plays the role of the transport cost. This separation generally comes at the price of an inequality, which we can explain as follows. In the optimal transport cost (5), mass is assigned to targets based on $j_t(s, z)$, which *averages* over future noise realizations. By contrast, the Bellman recursion (4) evaluates the optimal cost-to-go function $J_{t+1}(f_{t\#}\rho_t, \nu)$ at the realized future population $f_{t\#}\rho_t$. Thus, the optimal transport coupling is here computed *after* observing the noise realization. As optimal transport costs entail a minimization, they generally do not commute with an expectation. Interchanging them is the cause of the inequality (5). While the inequality could, in principle, be improved by parameterizing the single-agent cost to go in the noise realizations, this would lead to an intractable single-agent problem. Instead, averaging out the noise leads to stochastic optimal control problems and, thus, appears to be the best practically relevant solution. When the dynamics are deterministic, the next population state is fully determined by (μ_t, λ_t) and the expectation in (4) drop. As such, our upper bound in Theorem 3.1, Part I becomes exact, recovering the deterministic separation of [25].

Corollary 3.2 (Exact separation in the deterministic setting [25, Corollary 4.2]). *When the dynamics are deterministic, the population cost-to-go function (4) is an optimal transport cost with the single-agent cost-to-go function as its transportation cost:*

$$J_t(\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma(s, z).$$

From a bound to a policy. The proof of Theorem 3.1, Part I constructs an explicit population-level control $(\lambda_t^\varepsilon)_{t=0}^{T-1}$ that achieves cost within any $\varepsilon > 0$ of the bound. We denote $J^\lambda(\bar{\mu}_0, \nu)$ the expected cumulative cost of Problem 2 when the population follows the state-action distributions $(\lambda_t)_{t=0}^{T-1}$.

Theorem 3.1 continued (Separation principle, Part II: constructive separation-based control). *For each timestep t and every $\varepsilon > 0$, let γ_t^ε be an $\frac{\varepsilon}{2T}$ -optimal solution of the transport problem $\mathcal{K}[j_t](\mu_t, \nu)$ and let $a_t^\varepsilon : S \times Z \rightarrow A$ be an $\frac{\varepsilon}{2T}$ -optimal single-agent action for the Bellman recursion (3). Define the separation-based state-action distribution, $\lambda_t^\varepsilon := (\text{proj}_S^{S \times Z}, a_t^\varepsilon)_\# \gamma_t^\varepsilon$. Then,*

$$J^{\lambda^\varepsilon}(\bar{\mu}_0, \nu) \leq \inf_{\gamma \in \Gamma(\mu_0, \nu)} \int_{S \times Z} j_0(s, z) d\gamma(s, z) + \varepsilon.$$

In particular, if both an optimal γ_t^ and optimal single-agent actions a_t^* exist (e.g., all functions are continuous and all spaces compact), we can choose $\varepsilon = 0$.*

This construction has a natural interpretation. Assuming, for simplicity, existence of optimal couplings and actions, the coupling $\gamma_t^* \in \Gamma(\mu_t, \nu)$ is the optimal assignment of population mass to targets: for each state s , it specifies how the mass at s is assigned to the targets. The state-action distribution then assigns to each (s, z) pair the single-agent optimal action a_t^* that steers the agent from s to z . This gives rise to our separation principle: the coupling γ_t^* indicates how to dispatch agents to targets, while a_t^* indicates how to steer the single agent to the assigned target. Theorem 3.1 rigorously quantifies the performance of this construction, which is optimal in a deterministic setting.

Implication on agent-based formulation J^N . Theorem 3.1 is stated for the population formulation in Problem 2. Corollary 3.3 bridges this to the original agent-based setting: for N agents and N target masses, the same transport surrogate $\mathcal{K}[j_t]$ upper bounds J_t^N .

Algorithm 1 SALT: Separation-based Assignment and Learning via optimal Transport

Require: Initial target-conditioned policy $\pi^{(0)}(a \mid s, z)$, cost-to-go function estimate $\hat{j}^{(0)}(s, z)$, horizon T , exploration schedule $(\alpha)_{k \geq 0}$

- 1: **for** episodes $k = 0, 1, 2, \dots$ **do**
- 2: Sample the episode target distribution ν_k .
- 3: **Population-level matching.** With probability $1 - \alpha_k$, assign agents to targets via optimal transport with cost $\hat{j}^{(k)}$. Otherwise assign randomly.
- 4: **Single-agent learning.** Roll out the shared target-conditioned policy $\pi^{(k)}$ in parallel for all agents over the horizon T , with the assigned targets kept fixed.
- 5: Update $\pi^{(k+1)}$ and $\hat{j}^{(k+1)}(s, z)$ with the chosen single-agent RL algorithm from the N trajectories collected in Step 4.
- 6: **end for**
- 7: **Evaluation:**
- 8: **for** $t = 0, \dots, T - 1$ **do**
- 9: Assign agents to targets via optimal transport with cost $\hat{j}_t$.
- 10: Execute one step of π_t with the current target assignments.
- 11: **end for**

Corollary 3.3 (Agent-based upper bound). *Let $\mathbf{s}_t = (s_t^1, \dots, s_t^N) \in S^N$ be the agent states at time t , with empirical distribution $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{s_t^i}$. Let $\mathbf{z} = (z^1, \dots, z^N) \in Z^N$ be fixed targets with empirical distribution $\nu^N = \frac{1}{N} \sum_{i=1}^N \delta_{z^i}$. Then, the cost-to-go function for the optimal control problem in Problem 1, denoted by $J_t^N(\mathbf{s}_t, \nu^N)$, satisfies*

$$J_t^N(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N). \quad (6)$$

In particular, the inequality holds true for the cost to go $J^N(\bar{\mathbf{s}}_0, \nu^N)$ in Problem 1.

We defer the proof of Theorem 3.1, Part I, Theorem 3.1, Part II, and Corollary 3.3 to Appendix C.

4 SALT: Separation-based assignment and learning via optimal transport

Corollary 3.3 suggests using the upper bound

$$\mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int j_t(s, z) d\gamma(s, z)$$

as a tractable surrogate for the agent-based cost to go $J_t^N(\mathbf{s}_t, \nu^N)$. The resulting learning paradigm, summarized in Algorithm 1, has two phases: (i) learn the single-agent target-conditioned cost-to-go function $\hat{j}_t$, (ii) use it as the transport cost to assign agents to targets. The choice of single-agent algorithm is entirely open—any method capable of learning a target-conditioned cost-to-go estimate $\hat{j}_t$ (Q-learning [30], DDQN [31], PPO [32], A2C [33], etc.) can be plugged in as the base learner. We refer to this family of methods as SALT, for Separation-based Assignment and Learning via optimal Transport. An instance using a single-agent RL algorithm is denoted SALT-ALG; e.g., SALT-PPO.

Discussion. SALT replaces direct learning over the joint population state-action space with a target-conditioned single-agent policy $\pi(\cdot \mid s, z)$ and cost-to-go estimate $\hat{j}(s, z)$ shared across all agents, where the only agent-specific quantity is the target z assigned by the optimal transport step. This is desirable for three main reasons. (i) Scalability: Learning occurs entirely at the target-conditioned single-agent level, avoiding the curse of dimensionality of the joint state-action space and generalizing naturally across population sizes. (ii) Trajectory pooling: The same single-agent model is updated from every agent’s trajectory simultaneously. (iii) Modularity: The coordination layer requires only a cost-to-go estimate $\hat{j}(s, z)$ and, thus, SALT can be paired with any single-agent RL algorithm.

Training details. The surrogate $\mathcal{K}[j_t](\mu_t, \nu)$ requires a cost-to-go estimate $\hat{j}_t(s, z)$ used as the cost for population-level assignment, and a single-agent policy $\pi_t(a \mid s, z)$, executed once targets are assigned. Both are functions only of a single agent’s state and target, so learning is entirely at the single-agent level and independent of population size. At episode k , the initial distribution μ_0 and the

	Individual noise			Local noise			Global noise		
SALT-Q	24.2	29.4	39.5	24.2	29.4	39.6	24.2	29.0	38.9
SALT-DDQN	24.2	29.0	37.8	24.2	29.2	38.2	24.2	28.8	36.5
SALT-PPO	24.1	28.0	35.1	24.1	28.1	36.2	24.1	28.2	34.0
SALT-A2C	24.4	27.9	35.7	24.4	28.3	36.4	24.4	28.3	34.8
MAPPO	50.9	52.2	60.8	50.9	51.8	59.9	50.9	51.5	55.7
IPPO	47.4	52.5	59.8	47.4	50.1	59.0	47.4	50.2	54.6
QMIX	58.4	64.3	65.8	58.4	66.2	74.9	58.4	60.9	63.9
VDN	61.4	77.6	69.0	61.4	78.4	77.6	61.4	65.3	65.6
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 2: Average travel time to targets as a function of the noise probability p under individual, local, and global noise. Entries are means across 20 evaluation seeds. Brighter cells indicate worse relative performance. SALT variants consistently achieve lower travel times than standard MARL baselines.

target distribution ν (which is part of task specification) are sampled and agents are assigned to targets by solving the optimal transport problem using the current estimate $\hat{j}^{(k)}(s, z)$ as cost. For empirical measures with N samples, this optimal transport problem reduces to an assignment problem. It can be solved exactly in $O(N^3)$, or approximately using entropic regularization [34]. With probability α_k assignments are drawn uniformly at random instead, ensuring broad coverage of (s, z) pairs and preventing the cost-to-go estimate from overfitting to its own assignments. Assignments are kept fixed for the entire episode. This makes each agent’s problem a standard single-agent task with terminal cost $g_T(s_T, z^i)$, whose cost to go is exactly $j_t(s, z^i)$, and ensures that collected trajectories are consistent with the target-conditioned objective. Agents then roll out the shared target-conditioned policy and all transitions are pooled to update the shared policy/value model. Thus, a population of N agents provides N rollouts per episode covering diverse (s, z) pairs simultaneously. The precise update rule depends on the chosen single-agent RL algorithm. At deployment, the optimal transport assignment is recomputed at each timestep with cost $\hat{j}_t(s, z)$, and each agent acts under its newly assigned target. This mirrors the separation bound in Theorem 3.1, Part I and allows the policy to correct for stochastic deviations that accumulate over the episode.

5 Experimental results

We evaluate SALT in stochastic grid worlds, where we compare against MARL baselines, and in an urban logistic task in Manhattan road network, where we test scalability and transfer across population sizes. Code is at anonymous.4open.science/r/SALT-17FB/. More ablations and experiment details are in Section D.

5.1 Stochastic grid-world experiments

Separation leads to better performance. We first evaluate SALT on a stochastic grid-world environment. A population of $N = 16$ homogeneous agents must reach N target locations within a horizon $T = 100$. Agents start from the first column of an 11×30 grid, targets are uniformly sampled in the last ten columns. At each step, the intended action is replaced by a random one with probability p . We evaluate the three noise models introduced in Section 2: *individual*, *local* and *global* noise. The objective is to minimize average travel time and terminal mismatch to the target distribution. We compare four instances of SALT, using Q-learning [35] (SALT-Q), DDQN [31] (SALT-DDQN), PPO [32] (SALT-PPO), and A2C [33] (SALT-A2C) as single-agent routing algorithms, against four common deep MARL baselines, MAPPO [10], IPPO [10], VDN [12], QMIX [13]. All methods are trained and evaluated under the same task distribution and random seeds. Figure 2 shows that SALT consistently outperforms the MARL baselines across all noise levels and noise models. The four SALT variants perform similarly, suggesting that the gain comes from the separation structure rather than from the single-agent RL algorithm. Complementary metrics are reported in Appendix D.

Separation leads to better scalability. We next study how sample complexity scales with the

		SALT-PPO (ours)								MAPPO			
Reach-rate milestone	99%	3.8e6	7.7e7	5.2e7	2.9e7	8.5e7	1.8e8	4.5e8	1.5e9	—	—	—	—
	95%	8.9e5	2.2e6	3.5e6	6.0e6	1.8e7	6.0e7	2.3e8	8.6e8	1.1e7	3.0e8	1.6e9	—
	90%	6.3e5	1.8e6	3.0e6	5.2e6	1.5e7	5.1e7	1.9e8	7.3e8	9.3e6	2.4e8	1.4e9	—
	80%	4.7e5	1.4e6	2.5e6	4.3e6	1.3e7	4.3e7	1.6e8	6.3e8	8.9e6	2.2e8	1.2e9	—
	70%	3.9e5	1.2e6	2.2e6	3.8e6	1.1e7	3.8e7	1.5e8	5.6e8	6.7e6	1.9e8	9.7e8	—
		$N=2$ 10×10	$N=4$ 14×14	$N=6$ 17×17	$N=8$ 20×20	$N=16$ 28×28	$N=32$ 40×40	$N=64$ 57×57	$N=128$ 80×80	$N=2$ 10×10	$N=4$ 14×14	$N=6$ 17×17	$N=8$ 20×20
		Agents / grid size								Agents / grid size			

Figure 3: Training environment interactions for a reach-rate milestone (70% – 99%) for SALT-PPO and MAPPO, as a function of population size N (grid size scaled proportionally to keep target density constant). A dash (–) indicates the milestone was not reached within the $2e9$ -interaction budget.

number of agents. We use the same grid-world dynamics with individual noise, but vary the number of agents from $N = 2$ to $N = 128$. We keep the target density roughly constant to isolate the effects of scaling from the complexity of the task and avoid scenarios with highly condensed targets. Agents start at the bottom-left corner, and targets are sampled uniformly across the grid (see Appendix D.2). We select PPO as the single-agent learner in SALT, yielding SALT-PPO, and compare against MAPPO. We measure the number of environment interactions required to reach reach-rate milestones of 70%, 80%, 90%, 95%, and 99%, up to a maximum training budget of $2e9$ interactions. Figure 3 shows that SALT-PPO scales substantially better than MAPPO. SALT-PPO reaches high reach rates across all tested population sizes, while MAPPO scales poorly and often fails to reach the same milestones within the training budget. This scalability gap is directly explained by the separation principle. MAPPO must learn in the joint multi-agent state-action space, whose dimensionality grows with N , whereas SALT-PPO learns a single target-conditioned policy that is independent of N .

5.2 Towards a realistic time-varying stochastic urban logistics task

A fleet of delivery vehicles must serve requests distributed across Manhattan, jointly minimizing total travel time. This task couples vehicle-to-request assignment with time-efficient routing through a stochastic, time-varying network and is precisely the setting for which the separation principle was motivated. Rather than solving the full multi-agent problem online, SALT decomposes it into a coordination step (assigning vehicles to requests via optimal transport) and a routing step (navigating each vehicle to its target via a shared learned policy). The environment is a directed graph extracted from OpenStreetMap [36], containing the 2088 intersections and 4262 directed roads south of Central Park (see Figure 1). Travel times depend on road length, road type, traffic-light cycles of 90 seconds (with a road-dependent green phase duration and a random offset) and stochastic congestion effects. Further environment details are provided in Appendix D.3. A fleet of vehicles is initialized across three depot locations with random proportions and serves delivery requests sampled across the city, minimizing total travel time. We instantiate SALT with tabular Q-learning SALT-Q and compare it against an online vehicle-assignment baseline based on nominal shortest-path travel times. This baseline is a standard operational heuristic and ignores traffic-light timing and stochastic congestion, since these require spatiotemporal graphs whose size scales prohibitively with the horizon. The deep MARL baselines of Section 5.1 are excluded: as demonstrated, they become computationally prohibitive on a graph of this scale and are tied to a fixed training fleet size. All models are trained with $N = 20$ agents and evaluated, without retraining, on fleets of 1-100 vehicles.

Figure 1 reports the relative improvement in travel time as a function of fleet size. SALT-Q consistently outperforms the shortest-path baseline by roughly 20%, reflecting its ability to anticipate traffic-light timing and adapt to stochastic congestion. Performance remains stable across fleet sizes, confirming that SALT generalizes naturally to fleet sizes unseen at training time. The residual variance arises from a natural bias of distributional training: the policy concentrates on high-probability origin-destination pairs, improving sample efficiency at the cost of reduced accuracy on rare assignments.

6 Conclusion and limitations

We established a separation principle for cooperative MARL and used it to design SALT, a family of algorithms that learns a single-agent policy and uses optimal transport to coordinate agents. SALT outperforms existing MARL baselines in grid-worlds, both in terms of performance and scalability. In south Manhattan road network, where MARL baselines are impractical, SALT improves travel time by up to 20% over a shortest-path heuristic and generalizes to population sizes not seen during training.

Our work does not address several important challenges, which we leave to future work. From a theoretical perspective, our separation principle requires decoupled dynamics and an optimal transport terminal cost, ruling out settings with interactions, collisions, or non-transport objectives. Our bound on the optimal cost is also not tight in general. From a practical perspective, the target-conditioned single-agent problem must generalize across the target space, which is challenging when targets are high-dimensional or sampled from distributions far from the training one.

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A Noise models and averaging property

This appendix develops the measure-theoretic framework used to handle stochasticity in the multi-agent system. We first introduce the relevant general notation, and illustrate how common noise models fit into this framework through a series of examples. We conclude with Lemma A.9, which will be a key technical tool used in the proof of Theorem 3.1, Part I in Appendix C. Throughout, the space of probability measures on a Borel space X , $\mathcal{P}(X)$ is endowed with the topology of weak convergence and its Borel σ -algebra. Since X is a Polish space, $\mathcal{P}(X)$ is itself a Polish space [28, Theorem 15.15]. The product measure of two independent random variables with marginals $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is the unique measure $\mu \otimes \nu \in \mathcal{P}(X \times Y)$ [37, Theorem B, Section 35] such that $(\mu \otimes \nu)(A \times B) = \mu(A) \nu(B)$ for all Borel sets $A \subseteq X$, $B \subseteq Y$.

Table 1: Notation for the appendix.

Object	Definition	Intuition
$\hat{\xi}_t^{s,a}$	Random element of $\mathcal{P}(W)$	Distribution of noise experienced by agents in (s, a) at time t
$\eta_t^{s,a}$	Distribution of $\hat{\xi}_t^{s,a}$	Describes the probability distribution associated with the noise realizations
η_t	Probability measure in $\mathcal{P}(\mathcal{P}(W)^{S \times A})$	Collects all $\eta_t^{s,a}$ across bins, encoding population-level randomness
ρ_t	$\lambda_t \otimes \hat{\xi}_t^{s,a}$	Joint state-action-noise distribution

A.1 General framework

In multi-agent systems, randomness enters at the *individual* level, through the noise realizations for each individual agent. At the *population* level, these realizations induce randomness in the population distribution and therefore in the next population state. To capture this in a unified way, we work with distributions over noise distributions. Concretely, following the notation in Section 2.2, let $\lambda_t \in \mathcal{P}(S \times A)$ denote the joint state-action distribution of the population at time t , and denote with $\xi_t(\cdot | s, a) \in \mathcal{P}(W)$ the noise kernel, i.e., the law of the noise experienced by an agent in state s taking action a . For agents sharing the same state-action pair (s, a) , their noise realizations form an empirical distribution that we denote $\hat{\xi}_t^{s,a} \in \mathcal{P}(W)$. Unlike the kernel $\xi_t(\cdot | s, a)$, which is a fixed object, $\hat{\xi}_t^{s,a}$ is random—it fluctuates across episodes depending on the specific noise draws of the agents in that bin. Therefore, we can talk about the distribution of $\hat{\xi}_t^{s,a}$, that we denote by $\eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W))$, and collect these across all state-action pairs into the joint object $\eta_t \in \mathcal{P}(\mathcal{P}(W)^{S \times A})$. Finally, to describe the full population transition in this stochastic setting, it is not enough to track where agents are and what actions they take, one also needs to account for the noise they experience. This motivates the introduction of the joint state-action-noise distribution, defined as the gluing

$$\rho_t = \lambda_t(\mu_t) \otimes \hat{\xi}_t^{s,a} \in \mathcal{P}(S \times A \times W).$$

Intuitively, $\rho_t(\{s, a, w\})$ is the number of agents that are at state s , use action a , and experience the noise realization w . Since the noise realizations are random, $\hat{\xi}_t^{s,a}$ is random and, consequently, also ρ_t is random. At this point, the next population is a pushforward of μ under the system dynamics f_t :

$$\mu_{t+1} = f_{t\#} \rho_t.$$

Different noise models below correspond to different distributional assumptions on $\hat{\xi}_t^{s,a}$, and hence on ρ_t . The objects introduced above are summarized in Table 1.

We impose the following consistency condition on the realized noise model.

Assumption A.1. Fix a timestep t . There exists a Borel set $E_t \subset S \times A$ with $\lambda_t(E_t) = 1$ such that for all $(s, a) \in E_t$ and for every bounded measurable $\varphi : W \rightarrow \mathbb{R}$

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a). \quad (7)$$

Condition (7) should be read as an unbiased or barycenter condition between the population-level realized noise model $\hat{\xi}_t^{s,a}$ and the true underlying kernel $\xi_t(\cdot | s, a)$. More precisely, the assumption requires that these two descriptions agree after averaging over the realization of $\hat{\xi}_t^{s,a}$. As we will show below, Assumption A.1 is satisfied by several standard realized-kernel constructions. Specifically, In Proposition A.8 we show that it holds true for the noise models presented in Examples A.3- A.6. More generally, Assumption A.1 is satisfied whenever the realized kernel is chosen from a family of probability kernels whose mixture is the nominal kernel $\xi_t(\cdot | s, a)$.

What (7) excludes are biased realized-kernel constructions, as we show in the next example.

Example A.2. Let $W = \{0, 1\}$, $\xi_t = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. Suppose we artificially bias all noise realization towards 1, i.e., we have

$$\hat{\xi}_t^{s,a} = \delta_1$$

almost surely, then

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W w d\hat{\xi}_t^{s,a}(w) \right] = 1 \neq \frac{1}{2} = \int_W w d\xi_t(w | s, a).$$

Hence, (7) fails.

More generally, it fails whenever the realized population-level kernel systematically overweights some noise realizations relative to the nominal kernel used in the single-agent Bellman equation. In the remainder of this section, we specify this general framework to the individual, local, and global noise models considered in this work (Section A.2). Finally, in Section A.3, we prove a key property of all these noise models: we establish, in Lemma A.9, that the expected cost under the realized empirical kernel $\hat{\xi}_t^{s,a}$ equals the cost under the underlying kernel $\xi_t(\cdot | s, a)$. Ultimately, this unbiasedness property, verified for all four noise models in Proposition A.8, is what allows us to eliminate randomness from the backward recursion in Appendix C and recover the single-agent Bellman equation.

A.2 Noise models

To start, we consider the case where the noise is i.i.d. over all agents with the same state-action pair. We start with the case of finitely many agents:

Example A.3 (Individual noise, finitely many agents). *Consider N finite agents, each with its “own” noise sample*

$$w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i).$$

We assume noise draws to be independent across agents and across state-action pairs. Given the state-action distribution λ_t , its support $\text{supp } \lambda_t$ is finite (at most N atoms, since $\lambda_t = \frac{1}{N} \sum_{i=1}^N \delta_{(s_t^i, a_t^i)}$). For each bin $(s, a) \in \text{supp } \lambda_t$, let

$$N_{s,a} := N \cdot \lambda_t(\{(s, a)\}) \geq 1$$

be the number of agents in that bin. This is a strictly positive integer, determined by λ_t . Define the empirical measure map

$$\begin{aligned} \Psi_{N_{s,a}}^{s,a} : W^{N_{s,a}} &\rightarrow \mathcal{P}(W) \\ w^{s,a} &\mapsto \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \delta_{w_t^i}, \end{aligned}$$

where $w^{s,a} = (w_t^i)_{i \in [N_{s,a}]}$. The realized noise distribution for the state-action pair (s, a) is

$$\hat{\xi}_t^{s,a} = \Psi_{N_{s,a}}^{s,a}(w^{s,a}) \in \mathcal{P}(W),$$

with distribution

$$\eta_t^{s,a} = \Psi_{N_{s,a} \#}^{s,a} \left(\bigotimes_{i \in [N_{s,a}]} \xi_t(\cdot | s, a) \right) \in \mathcal{P}(\mathcal{P}(W)).$$

Since $\text{supp } \lambda_t$ is finite and noise realizations are independent across state-action pairs, the joint law is the finite product

$$\eta_t = \bigotimes_{(s,a) \in \text{supp } \lambda_t} \eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W)^{\text{supp } \lambda_t}).$$

We extend η_t to all state-action pairs by setting, for $(s, a) \notin \text{supp } \lambda_t$, $\hat{\xi}_t^{s,a} = \delta_{w^0}$, for an arbitrarily chosen $w^0 \in W$. Clearly, the specific choice of w^0 is inconsequential: since λ_t assigns zero mass to these bins, they do not contribute λ_t -weighted integrals—exactly the type of integrals of our problem setting. The map $(s, a) \mapsto \eta_t^{s,a}$ is Borel measurable: it takes finitely many distinct values (one per element of $\text{supp } \lambda_t$, plus the constant extension outside), and the preimage of each value is a finite union of singletons or the complement of $\text{supp } \lambda_t$, all of which are Borel sets.

Example A.4 (Individual noise, infinitely many agents). *In the limit case of infinitely many agents, the two-level randomization—each agent drawing a private noise sample and the collection of noise realizations across the population—collapse into a single draw from the deterministic coupling*

$$\rho_t = \lambda_t \otimes \xi_t(\cdot | s, a).$$

Intuitively, the population is “so vast” that its empirical profile of noise realizations matches the underlying noise kernel itself. Rigorously, in the case of finite state and action spaces, we can assume that the number of agents in each state-action pair (s, a) grows to infinity and Varadarajan’s theorem [38, Theorem 11.4.1.] in the space $\mathcal{P}(W)$ guarantees

$$\hat{\xi}_t^{s,a} \xrightarrow{a.s.} \xi_t, \quad \text{as } N_{s,a} \rightarrow \infty.$$

Consequently, the laws of the realized empirical kernels $\eta_t^{s,a}$ converges weakly to $\delta_{\xi_t(\cdot | s, a)}$ [38, Theorem 9.2.1, Proposition 9.3.5]. We therefore set $\hat{\xi}_t^{s,a} = \xi_t(\cdot | s, a)$ almost surely and

$$\eta_t^{s,a} = \delta_{\xi_t(\cdot | s, a)} \in \mathcal{P}(\mathcal{P}(W)).$$

For general (possibly uncountable) Polish spaces S and A , we construct the joint law via Kolmogorov’s extension theorem [39, Theorems 36.1 and 36.2]. Specifically, for each finite subset $J \subset S \times A$, define a probability measure

$$\eta_t^J = \bigotimes_{(s,a) \in J} \delta_{\xi_t(\cdot | s, a)} \quad \text{on } \mathcal{P}(\mathcal{P}(W))^J.$$

These finite-dimensional marginals are consistent (the marginal of η_t^J onto any $J' \subset J$ equals $\eta_t^{J'}$, since product of Dirac masses marginalize correctly). Thus, Kolmogorov’s extension theorem guarantees that there exists a unique probability measure

$$\eta_t = \bigotimes_{(s,a) \in S \times A} \delta_{\xi_t(\cdot | s, a)},$$

and the realized joint state-action-noise distribution collapses to the deterministic coupling $\rho_t = \lambda_t \otimes \xi_t(\cdot | s, a)$.

Example A.5 (Local noise). *Suppose the noise realizations are shared among agents with the same state-action pair. For each fixed $(s, a) \in S \times A$ we draw*

$$w_t^{s,a} \sim \xi_t(\cdot | s, a),$$

independently across state-action pairs. Then, all agents in bin (s, a) experience the same noise realization $w_t^i = w_t^{s,a}$. In this framework, we define the empirical measure map

$$\begin{aligned} \Psi^{s,a}: W &\rightarrow \mathcal{P}(W) \\ w_t^{s,a} &\mapsto \delta_{w_t^{s,a}}. \end{aligned}$$

The realized noise distribution for bin (s, a) collapses to

$$\hat{\xi}_t^{s,a} = \delta_{w_t^{s,a}},$$

with distribution

$$\eta_t^{s,a} = \Psi^{s,a} \# \xi_t(\cdot | s, a) \in \mathcal{P}(\mathcal{P}(W)).$$

Since noise samples are independent across state-action pairs, the joint law is constructed by the same Kolmogorov extension argument as in Example A.4 (applied with $\eta_t^{s,a}$ in place of $\delta_{\xi_t(\cdot | s, a)}$). For every finite $J \subset S \times A$, the marginal $\eta_t^J = \bigotimes_{(s,a) \in J} \eta_t^{s,a}$ is consistent, yielding the unique extension

$$\eta_t = \bigotimes_{(s,a) \in S \times A} \eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W)^{S \times A}).$$

Example A.6 (Global noise). *Consider the case in which all agents are subject to the same noise draw $w_t \sim \xi_t \in \mathcal{P}(W)$, regardless of their state and action. Since every state-action pair (s, a) experiences the same noise realization, we have $\hat{\xi}_t^{s,a} = \delta_{w_t}$ for all (s, a) . We define then the diagonal embedding*

$$\begin{aligned} \Phi: W &\rightarrow \mathcal{P}(W)^{S \times A} \\ w &\mapsto ((s, a) \mapsto \delta_w). \end{aligned}$$

The entire random family $(\hat{\xi}_t^{s,a})_{(s,a) \in S \times A} = \Phi(w_t)$ is a $\mathcal{P}(W)^{S \times A}$ -valued random variable and its law is

$$\eta_t = \Phi_{\#} \xi_t \in \mathcal{P}(\mathcal{P}(W)^{S \times A}).$$

In words, $\xi_t \in \mathcal{P}(W)$ is pushed forward through Φ to a measure on $\mathcal{P}(W)^{S \times A}$, concentrated on the diagonal $\{\Phi(w) : w \in W\}$. We refer the reader to Remark A.7 for a discussion on the well-posedness of Φ .

Remark A.7 (Well-posedness of Example A.6). We verify here that the map $\Phi : W \rightarrow \mathcal{P}(W)^{S \times A}$ is well-defined and a Borel measurable map between Polish spaces, so that the pushforward $\eta_t = \Phi_{\#} \xi_t$ is well-defined.

Step 1: Each coordinate map is continuous. We start by noting that each coordinate map $w \mapsto \delta_w : W \rightarrow \mathcal{P}(W)$ is continuous with respect to the weak topology on $\mathcal{P}(W)$ [39, Example 2.1].

Step 2: Φ is Borel measurable. To show that the full map Φ is Borel measurable we consider two cases depending on whether $S \times A$ is countable or uncountable. In the countable case, $\mathcal{P}(W)^{S \times A}$ is a countable product of Polish spaces, which is again a Polish space [40, Proposition 3.3] under the product topology. A map into a countable product is measurable if and only if each coordinate projection composed with it is measurable, which holds by Step 1. So Φ is Borel measurable. If, instead, $S \times A$ is uncountable, then $\mathcal{P}(W)^{S \times A}$ might fail to be Polish. However, this does not pose issues as the image of Φ is contained in the diagonal subset

$$\mathcal{D} = \{\Phi(w) : w \in W\} = \{((s, a) \mapsto \delta_w) : w \in W\} \subset \mathcal{P}(W)^{S \times A}.$$

The map $\Phi : W \rightarrow \mathcal{D}$ is a homeomorphism: both Φ and its inverse $\Phi^{-1} : \mathcal{D} \rightarrow W$ (which evaluates any coordinate and then inverts $w \mapsto \delta_w$) are continuous. Hence, $\mathcal{D}$ is homeomorphic to W , which is Polish. In particular, $\mathcal{D}$ is a Polish space in the subspace topology inherited from $\mathcal{P}(W)^{S \times A}$. Since $\Phi : W \rightarrow \mathcal{D}$ is continuous between Polish spaces it is Borel measurable, and $\eta_t = \Phi_{\#} \xi_t$ is a well-defined Borel probability measure on $\mathcal{D}$ and so also on $\mathcal{P}(W)^{S \times A}$. We finally observe that η_t is fully concentrated on $\mathcal{D}$, so all subsequent integrals only involve elements of $\mathcal{D}$ and the uncountability of $S \times A$ does not pose an issue.

We now show that these canonical noise models satisfy Assumption A.1.

Proposition A.8 (The noise models satisfy Assumption A.1). *For each of the noise models in Examples A.3-A.6, the realized kernel $\hat{\xi}_t^{s,a}$ and its distribution η_t satisfy Assumption A.1. That is, there exists a Borel set $E_t \in S \times A$ with $\lambda_t(E_t) = 1$ such that for all $(s, a) \in E_t$ and for every bounded measurable $\varphi : W \rightarrow \mathbb{R}$ simultaneously,*

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Proof. We prove the statement for each noise model separately.

Individual noise (Example A.3). Fix $(s, a) \in \text{supp } \lambda_t$, hence $N_{s,a} \geq 1$. By construction,

$$\hat{\xi}_t^{s,a} = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \delta_{w_t^i}$$

where w_t^i are i.i.d with law $\xi_t(\cdot | s, a)$. Hence, for any bounded measurable $\varphi : W \rightarrow \mathbb{R}$,

$$\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \varphi(w_t^i)$$

and therefore, by linearity of expectation,

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \mathbb{E}_{\hat{\xi}_t^{s,a}} [\varphi(w_t^i)] = \int_W \varphi(w) d\xi_t(w | s, a).$$

thus (7) holds on $E_t = \text{supp } \lambda_t$, which clearly satisfies $\lambda_t(E_t) = 1$.

Infinite-population model (Example A.4). By construction $\hat{\xi}_t^{s,a} = \xi_t(\cdot | s, a)$ almost surely for every $(s, a) \in S \times A$. Therefore, for every bounded measurable φ

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Thus, (7) holds with $E_t = S \times A$.

Local noise (Example A.5). For each fixed (s, a) , one draw $w_t^{s,a} \sim \xi_t(\cdot | s, a)$ independently across state-action pairs, and the realized noise distribution is $\hat{\xi}_t^{s,a} = \delta_{w_t^{s,a}}$. So, for every bounded measurable φ

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \mathbb{E}_{\hat{\xi}_t^{s,a}} [\varphi(w_t^{s,a})] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Thus, (7) holds again with $E_t = S \times A$.

Global-noise (Example A.6). This case is identical to the local noise one, with the same draw w_t shared across all (s, a) . $\square$

A.3 Averaging property of realized noise kernels

We now show that even if $\hat{\xi}_t^{s,a}$ is random, the expected cost computed under its law η_t equals the cost computed directly under the underlying kernel $\xi_t(\cdot | s, a)$. In other words, replacing the random empirical noise law $\hat{\xi}_t^{s,a}$ by the underlying kernel $\xi_t(\cdot | s, a)$ is unbiased in expectation. Among others, this lemma will allow us to remove randomness from the backward recursion in the proof of Theorem 3.1, Part I.

Lemma A.9 (Averaging realized kernels). *Let S, A, W be Polish spaces and fix a timestep t . Let λ_t denote the joint state-action distribution and $\xi_t : S \times A \rightarrow \mathcal{P}(W)$ be a stochastic kernel. For each (s, a) , let $\hat{\xi}_t^{s,a}$ be a $\mathcal{P}(W)$ -valued random variable with distribution $\eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W))$, so that Assumption A.1 holds true and $(s, a) \mapsto \eta_t^{s,a}$ is measurable. Then, for every nonnegative measurable $c : S \times A \times W \rightarrow \mathbb{R}_{\geq 0}$,*

$$\int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] d\lambda_t(s, a) = \int_{S \times A} \int_W c(s, a, w) d\xi_t(w | s, a) d\lambda_t(s, a). \quad (8)$$

Proof. For $n \in \mathbb{N}$, define

$$c_n(s, a, w) := \min\{c(s, a, w), n\}.$$

By construction, each c_n is bounded and measurable, and $c_n \uparrow c$ pointwise.

To start, we prove that the two integrands in (8) are measurable. For the left-hand side, for each n , the map $(s, a, \zeta) \mapsto \int_W c_n d\zeta$ is Borel measurable on $S \times A \times \mathcal{P}(W)$ and continuous in ζ . This follows from a standard monotone-class argument, starting from indicators of measurable rectangles and using the fact that $\zeta \mapsto \zeta(B)$ is Borel measurable for every Borel set $B \in W$ [29, Proposition 7.28]. Since $(s, a) \mapsto \eta_t^{s,a}$ is measurable, standard measurability arguments of the stochastic kernel yield measurability

$$(s, a) \mapsto \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right].$$

Letting $n \uparrow \infty$, the monotone convergence theorem [39, Theorem 16.2] yields the measurability of

$$(s, a) \mapsto \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c(s, a, w) d\hat{\xi}_t^{s,a}(w) \right].$$

For the right-hand side, instead, the measurability of

$$(s, a) \mapsto \int_W c(s, a, w) d\xi_t(w | s, a).$$

follows from standard properties of stochastic kernels [29, Propositions 7.28 and 7.29].

We now prove the equality (8). Fix n , for every $(s, a) \in E_t$, apply (7) with the bounded measurable test function $\varphi_{s,a}(w) := c_n(s, a, w)$, viewed as a function of w alone for fixed (s, a) . This gives

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W c_n(s, a, w) d\xi_t(w | s, a)$$

for all $(s, a) \in E_t$. Since $\lambda_t(E_t) = 1$, integrating both sides with respect to λ_t gives

$$\int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] d\lambda_t(s, a) = \int_{S \times A} \int_W c_n(s, a, w) d\xi_t(w | s, a) d\lambda_t(s, a).$$

Finally, let $n \rightarrow \infty$. Since $c_n \uparrow c$, the monotone convergence theorem yields (8). $\square$

B Preliminary lemmata

We now collect the auxiliary results required for the main proof in Appendix C.

Measurability convention. Throughout the appendix, primitive data such as the dynamics, stage costs, and terminal cost are assumed to be Borel measurable, as in the main text. Since every nonnegative Borel measurable function is lower semianalytic, these data also satisfy the analytic-measurability assumptions used in the auxiliary results below. We use lower semianalytic functions because the dynamic programming recursion involves pointwise infima over the action space, and such infima need not be Borel measurable on general Polish spaces; see [29] for a detailed discussion. Accordingly, value functions and pointwise Bellman infima are interpreted as lower semianalytic, hence universally measurable, functions. This is sufficient for the integrals and measurable-selection arguments used in the proof of Theorem 3.1. We work at this level of generality to simultaneously capture systems with finite and continuous state and actions spaces. Nevertheless, when all spaces are finite, measurability is automatic and all measurability arguments can therefore be skipped.

We begin by recalling from [25] a result that lets us “move” pushforward from the set of coupling into the transportation cost, and vice versa.

Lemma B.1 (Pushforward and optimal transport [25, Lemma 6.1]). *Given a transportation cost $j_{t+1} : S \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$, a map $f_t : S \times A \times W \rightarrow S$, and probability measure $\rho_t \in \mathcal{P}(S \times A \times W)$, it holds that*

$$\inf_{(\text{proj}_{S \times A \times W}^{S \times A \times W \times Z})_{\#} \gamma = \rho_t} \int_{S \times A \times W \times Z} j_{t+1} \circ (f_t \times \text{Id}_Z) d\gamma = \inf_{(\text{proj}_S^{S \times Z})_{\#} \tilde{\gamma} = f_{t\#} \rho_t} \int_{S \times Z} j_{t+1} d\tilde{\gamma},$$

where Id_Z the identity map over Z .

In our setting, this lemma tells us that evaluating the cost after applying the system dynamics is equivalent to pushing forward the distribution first. Thus, this means we can “apply the dynamics first” and then evaluate the population-level cost without changing the result.

Next, we recall how to merge additive costs into a single transport objective.

Lemma B.2 (Sum of optimal transport discrepancies [25, Lemma 6.2]). *Given transportation costs $c_1 : X \rightarrow \bar{\mathbb{R}}_{\geq 0}$, $c_2 : X \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$, and probability measures $\rho \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\mathbb{E}_{\rho} [c_1] + \mathcal{K}[c_2](\rho, \nu) = \mathcal{K}[c](\rho, \nu),$$

with $c : X \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$ defined as $c(x, z) = c_1(x) + c_2(x, z)$.

This result allows us to collapse stage costs and terminal transport costs into a unified expression, which is crucial for dynamic programming formulations.

We now introduce stochasticity which, in our setting, will appear due to the stochastic dynamics. Through a standard argument, we show that we can safely exchange expectation over random noise with the infimum over couplings, at the price of an inequality.

Lemma B.3 (Infimum and expectation). *Let $\lambda \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Z)$. Consider $\hat{\xi} = (\hat{\xi}^x)_{\{x \in X\}} \in \mathcal{P}(W)^X$ be a random family with $\hat{\xi}^x \in \mathcal{P}(W)$, and denote with $\eta \in \mathcal{P}(\mathcal{P}(W)^X)$ its distribution. Let $c : X \times W \times Z \rightarrow \mathbb{R}_{\geq 0}$ be measurable. Then*

$$\mathbb{E}_{\hat{\xi}} \left[\inf_{\gamma \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \right] \leq \inf_{\gamma \in \Gamma(\lambda, \nu)} \mathbb{E}_{\hat{\xi}} \left[\int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \right].$$

Proof. Fix any $\gamma \in \Gamma(\lambda, \nu)$. Pointwise in $\hat{\xi}$,

$$\inf_{\gamma \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \leq \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z).$$

Taking expectations over $\hat{\xi}$ preserves the inequality. Since the right-hand side holds for every coupling γ , taking the infimum over $\Gamma(\lambda, \nu)$ yields the claim. $\square$

With this inequality, randomness can be “pushed inside” the transport problem, a step that will later allow us to handle noisy transitions.

The next tool, recalled from [25], refines our handling of couplings: it shows how an optimizer can be disintegrated with respect to one marginal, yielding conditional distributions.

Lemma B.4 (Disintegration of the optimizer [25, Lemma 6.4]). *Given a transportation cost $c : S \times A \times Z \rightarrow \mathbb{R}_{\geq 0}$ and probability measures $\mu \in \mathcal{P}(S)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\begin{aligned} \inf_{\substack{(\text{proj}_{S \times Z}^{S \times A \times Z})_{\#} \\ \gamma \in \Gamma(\mu, \nu)}} \int_{S \times A \times Z} c(s, a, z) d\gamma(s, a, z) \\ = \inf_{\gamma' \in \Gamma(\mu, \nu)} \inf_{\{\alpha^{s,z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma'(s, z), \end{aligned} \quad (9)$$

where $\Lambda(A) := \{\{\alpha^{s,z}\}_{(s,z) \in S \times Z} \subseteq \mathcal{P}(A) \mid S \times Z \ni (s, z) \mapsto \alpha^{s,z} \in \mathcal{P}(A) \text{ Borel}\}$.

This separation will allow us to express the population-level policy as a distribution over conditional agent-level actions.

Next, we study more in detail the right-hand side in (9), showing that the infimum over the family $\{\alpha^{s,z}\} \in \Lambda(A)$ can be “pushed” into the integral.

Lemma B.5 (Reduction to pointwise infimum over deterministic policies). *Let S, A, Z be Polish spaces and let $c : S \times A \times Z \rightarrow \mathbb{R}_{\geq 0}$. Let $\mu \in \mathcal{P}(S)$, $\nu \in \mathcal{P}(Z)$ and $\gamma \in \Gamma(\mu, \nu)$. Define*

$$\Lambda(A) := \{\{\alpha^{s,z}\}_{(s,z) \in S \times Z} \subseteq \mathcal{P}(A) \mid S \times Z \ni (s, z) \mapsto \alpha^{s,z} \in \mathcal{P}(A) \text{ is Borel measurable}\}.$$

Then, the function $(s, z) \mapsto \inf_{a \in A} c(s, a, z)$ is universally measurable, and for every coupling $\gamma \in \Gamma(\mu, \nu)$,

$$\inf_{\alpha^{s,z} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma(s, z) = \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z). \quad (10)$$

Proof. We start by assessing the measurability of the pointwise infimum. Since c is lower semianalytic (see the measurability convention at the beginning of Appendix B) and A is a Polish space, the function $(s, z) \mapsto \inf_{a \in A} c(s, a, z)$ is lower semianalytic by [29, Proposition 7.47], hence universally measurable. In particular, the right-hand side of (10) is a well-defined element of $[0, +\infty]$.

We then prove (10) by showing “ $\leq$ ” and “ $\geq$ ” separately.

Lower bound “ $\geq$ ”. For any $\{\alpha^{s,z}\} \in \Lambda(A)$ and every $(s, z) \in S \times Z$, since $\alpha^{s,z} \in \mathcal{P}(A)$ is a probability measure

$$\int_A c(s, a, z) d\alpha^{s,z}(a) \geq \inf_{a \in A} c(s, a, z).$$

Integrating both sides with respect to γ and taking the infimum over all $\{\alpha^{s,z}\} \in \Lambda(A)$ leads to

$$\inf_{\{\alpha^{s,z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma(s, z) \geq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z).$$

Upper bound “ $\leq$ ”. Let $\varepsilon > 0$. Since $c \geq 0$ is lower semianalytic and A is a Polish space, [29, Proposition 7.50] guarantees the existence of a universally measurable ε -optimal selector $a^\varepsilon : S \times Z \rightarrow A$ satisfying

$$c(s, a^\varepsilon(s, z), z) \leq \inf_{a \in A} c(s, a, z) + \varepsilon \quad \forall (s, z) \in S \times Z.$$

Now, since γ is a σ -finite measure on $S \times Z$, by [29, Lemma 7.27] there exists a Borel measurable map $b^\varepsilon : S \times Z \rightarrow A$ with $b^\varepsilon = a^\varepsilon$ γ -almost everywhere, so that

$$c(s, b^\varepsilon(s, z), z) \leq \inf_{a \in A} c(s, a, z) + \varepsilon \quad \gamma\text{-almost everywhere.}$$

The Dirac embedding $A \rightarrow \mathcal{P}(A)$ defined by $a \mapsto \delta_a$ is continuous in the weak topology [39, Example 2.1], hence Borel measurable. Therefore, the composition $(s, z) \mapsto \delta_{b^\varepsilon(s, z)}$ is Borel measurable, and so is the family $\{\delta_{b^\varepsilon(s, z)}\}_{(s, z) \in S \times Z} \in \Lambda(A)$. Thus,

$$\begin{aligned} \inf_{\{\alpha^{s,z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma(s, z) &\leq \int_{S \times Z} c(s, b^\varepsilon(s, z), z) d\gamma(s, z) \\ &\leq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we conclude

$$\inf_{\{\alpha^{s,z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma(s, z) \leq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z).$$

This completes the proof of the upper bound and, with it, the proof of the lemma. $\square$

This lemma will allow us to relate the dynamic programming over probability measures with the standard dynamic programming over the state and action spaces.

Finally, we present a lemma that upper bounds the transport cost when replacing “full couplings” with “marginal plans”. This inequality will be used to simplify the optimization by relaxing the coupling constraints, enabling us to separate noise from the population-level allocation.

Lemma B.6 (Suboptimality of marginal plans). *Given a transportation cost $c : X \times W \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$ and probability measures $\rho \in \mathcal{P}(X \times W)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{X \times W \times Z} c(x, w, z) d\gamma(x, w, z) \leq \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z), \quad (11)$$

where $\rho = \lambda \otimes \hat{\xi}^x$ is a disintegration of ρ with respect to its first marginal λ .

Proof. Consider the disintegration of ρ with respect to its first marginal $\lambda = (\text{proj}_X^{X \times W})_\# \rho \in \mathcal{P}(X)$, $\rho = \lambda \otimes \hat{\xi}^x$. Let $\gamma_1 \in \Gamma(\lambda, \nu)$ and consider the gluing $\bar{\gamma} = \gamma_1 \otimes \hat{\xi}^x \in \mathcal{P}(X \times Z \times W)$. To “reorder” marginals, consider the permutation

$$\sigma : X \times Z \times W \rightarrow X \times W \times Z, \quad \sigma(x, z, w) = (x, w, z),$$

and set $\gamma = \sigma_\#(\gamma_1 \otimes \hat{\xi}^x) \in \mathcal{P}(X \times W \times Z)$.

We now show that γ is a coupling between ρ and ν and, thus, a valid upper bound for the infimum in (11). We check each marginal by testing against an arbitrary bounded measurable function. Let

$\varphi: X \times W \rightarrow \mathbb{R}$ be bounded and measurable, we have

$$\begin{aligned}
& \int_{X \times W} \varphi(x, w) d \left(\left(\text{proj}_{X \times W}^{X \times W \times Z} \right)_{\#} \gamma \right) (x, w) \\
&= \int_{X \times W \times Z} \varphi \left(\text{proj}_{X \times W}^{X \times W \times Z} (x, w, z) \right) d\gamma(x, w, z) \\
&= \int_{X \times W \times Z} \varphi \left(\text{proj}_{X \times W}^{X \times W \times Z} (x, w, z) \right) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\
&= \int_{X \times Z \times W} \varphi \left(\text{proj}_{X \times W}^{X \times W \times Z} (\sigma(x, z, w)) \right) d \left(\gamma_1 \otimes \hat{\xi}^x \right) (x, z, w) \\
&= \int_{X \times Z} \int_W \varphi(x, w) d\hat{\xi}^x(w) d\gamma_1(x, z) \\
&= \int_X \int_W \varphi(x, w) d\hat{\xi}^x(w) d\lambda(x).
\end{aligned}$$

For every bounded and measurable φ , since $\hat{\xi}^x$ is the disintegration of ρ on W given x , and $\gamma_1 \in \Gamma(\lambda, \nu)$ has first marginal λ , Fubini–Tonelli [39, Theorem 13.8] gives

$$\int_X \int_W \varphi(x, w) d\hat{\xi}^x(w) d\lambda(x) = \int_{X \times W} \varphi(x, w) d \left(\lambda \otimes \hat{\xi}^x \right) (x, w) = \int_{X \times W} \varphi(x, w) d\rho(x, w).$$

Hence $(\text{proj}_{X \times W}^{X \times W \times Z})_{\#} \gamma = \rho$. Similarly, for the marginal over Z , let $\psi: Z \rightarrow \mathbb{R}$ be bounded and measurable. Then,

$$\begin{aligned}
& \int_Z \psi(z) d \left(\left(\text{proj}_Z^{X \times W \times Z} \right)_{\#} \gamma \right) (z) \\
&= \int_{X \times W \times Z} \psi \left(\text{proj}_Z^{X \times W \times Z} (x, w, z) \right) d\gamma(x, w, z) \\
&= \int_{X \times W \times Z} \psi \left(\text{proj}_Z^{X \times W \times Z} (x, w, z) \right) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\
&= \int_{X \times Z \times W} \psi \left(\text{proj}_Z^{X \times W \times Z} (\sigma(x, z, w)) \right) d \left(\gamma_1 \otimes \hat{\xi}^x \right) (x, z, w) \\
&= \int_{X \times Z} \int_W \psi(z) d\hat{\xi}^x(w) d\gamma_1(x, z).
\end{aligned}$$

Since $\int_W d\hat{\xi}^x(w) = 1$ for all x and ν is the marginal over Z of $\gamma_1 \in \Gamma(\lambda, \nu)$, this reduces to

$$\int_{X \times Z} \psi(z) d\gamma_1(x, z) = \int_Z \psi(z) d \left(\left(\text{proj}_Z^{X \times W \times Z} \right)_{\#} \gamma \right) (z) = \int_Z \psi(z) d\nu(z),$$

Hence $(\text{proj}_Z^{X \times W \times Z})_{\#} \gamma = \nu$. Overall, we have proven that, for all $\gamma_1 \in \Gamma(\lambda, \nu)$, $\gamma = \sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \in \Gamma(\rho, \nu)$.

We are now ready to prove (11). Fix $\gamma_1 \in \Gamma(\lambda, \nu)$. Since the coupling $\gamma = \sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \in \Gamma(\rho, \nu)$ is a valid candidate for the left-hand side. Thus, we obtain the upper bound

$$\begin{aligned}
\inf_{\tilde{\gamma} \in \Gamma(\rho, \nu)} \int_{X \times W \times Z} c(x, w, z) d\tilde{\gamma}(x, w, z) &\leq \int_{X \times W \times Z} c(x, w, z) d\gamma(x, w, z) \\
&= \int_{X \times W \times Z} c(x, w, z) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\
&= \int_{X \times Z \times W} c(x, w, z) d \left(\gamma_1 \otimes \hat{\xi}^x \right) (x, z, w) \\
&= \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z),
\end{aligned}$$

where the last equality uses disintegration and Tonelli's theorem to get. Since $\gamma_1 \in \Gamma(\lambda, \nu)$ is arbitrary, we may take infima to obtain

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int c(x, w, z) d\gamma(x, w, z) \leq \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z),$$

which is the desired statement. $\square$

Through the following example, we show that the inequality (11) is generally strict.

Example B.7 (Strictness of Lemma B.6). *Let $X = W = Z = \mathbb{R}$. Set $\lambda = \mathcal{N}(0, 1)$ on X . For each $x \in X$, let $\hat{\xi} = \hat{\xi}^x = \mathcal{N}(0, 1)$ on W , independent of x . Then, the product measure $\rho = \lambda \otimes \hat{\xi}$ is the law of a pair (x, w) with $x, w \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$. Let $\nu = \mathcal{N}(0, 1)$ on Z and define the cost*

$$c(x, w, z) = (w - z)^2.$$

To evaluate the left-hand side in (11) in Lemma B.6, we aim to solve

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{\mathbb{R}^3} c(x, w, z) d\gamma(x, w, z).$$

To do so, we show that there exists a plan $\tilde{\gamma} \in \Gamma(\rho, \nu)$ that attains a cost of 0, which, by non-negativity of the cost, implies optimality. Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, w) = (x, w, w)$ and set $\tilde{\gamma} = T_{\#}\rho$. By construction, its (x, w) -marginal is ρ and its z -marginal is given by $w \sim \mathcal{N}(0, 1) = \nu$, hence $\tilde{\gamma} \in \Gamma(\rho, \nu)$. Under this coupling

$$\int_{\mathbb{R}^3} (w - z)^2 d\tilde{\gamma}(x, w, z) = \int_{\mathbb{R}^3} (w - w)^2 d\rho(x, w) = 0$$

and so

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{\mathbb{R}^3} c(x, w, z) d\gamma(x, w, z) = 0.$$

We now consider the right-hand side in (11). For any $\gamma_1 \in \Gamma(\lambda, \nu)$, we evaluate

$$\int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z).$$

Since the disintegration of ρ is $\hat{\xi} = \mathcal{N}(0, 1)$ (in particular, it is independent of x), we have

$$\int_{\mathbb{R}} c(x, w, z) d\hat{\xi}(w) = \int_{\mathbb{R}} (w - z)^2 d\hat{\xi}(w) = \int_{\mathbb{R}} w^2 d\hat{\xi}(w) - 2z \int_{\mathbb{R}} w d\hat{\xi}(w) + z^2 = 1 + z^2,$$

and so

$$\begin{aligned} \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{\mathbb{R}^2} \int_{\mathbb{R}} c(x, w, z) d\hat{\xi}(w) d\gamma_1(x, z) &= \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{\mathbb{R}^2} 1 + z^2 d\gamma_1(x, z) \\ &= 1 + \int_{\mathbb{R}} z^2 d\nu(z) \\ &= 2, \end{aligned}$$

where the last equality follows from the fact that the second marginal of γ_1 is $\nu = \mathcal{N}(0, 1)$ and thus its second moment is 1. Clearly, the left- and right hand sides do not agree, and the inequality (11) is strict in general.

Example B.7 shows that the inequality of Lemma B.6—and hence the bound of Theorem 3.1, Part I—is generally strict. Informally, the gap corresponds to the value of being able to match noise realizations directly to targets, rather than averaging over them. In control problems where noise is small relative to the nominal cost (as in the low-noise regime of our grid-world experiments), we expect this gap to be small.

C Proofs of Section 3

We now prove the main results presented in Section 3. Recall the single-agent problem and the dynamic programming for the cost-to-go function in (2):

$$\begin{aligned} j_T(s_T, z) &= g_T(s_T, z); \\ j_t(s_t, z) &= \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s_t, a_t, w_t) + j_{t+1}(f_t(s_t, a_t, w_t), z)]. \end{aligned}$$

C.1 Proof of Theorem 3.1

We aim to show that the population level cost-to-go function in Problem 2 admits as upper bound an optimal transport problem whose cost is the single-agent cost-to-go function in (2). Concretely, we show by backward induction that

$$J_t(\mu_t, \nu) \leq \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int j_t(s, z) d\gamma(s, z).$$

We prove the theorem under the more general setting where Assumption A.1 is in place. As shown in Proposition A.8, this assumption is satisfied by all three noise models—individual, local, and global—considered in the main body of the paper. This will allow us to use Lemma A.9 in the proof of the theorem.

We start with the proof of Part I of the theorem.

Proof. The proof proceeds by induction, in three steps. First, we use the definition of the cost-to-go function and the induction hypothesis at time $t + 1$. Second, we express the conditional expectations over noise realizations in terms of the averaged single-agent cost. Finally, we reorganize the infima over couplings by means of Lemma B.4, which allows us to recover the desired distributional dynamic program and conclude the induction.

The base case of the induction is straightforward. By definition $J_T(\mu_T) = \mathcal{K}[g_T](\mu_T, \nu)$. Since $j_T(s, z) = g_T(s, z)$, we have $\mathcal{K}[j_T](\mu_T, \nu) = \mathcal{K}[g_T](\mu_T, \nu) = J_T(\mu_T)$. Fix $t < T$, using the induction hypothesis at time $t + 1$, i.e.,

$$J_{t+1}(f_{t\#}\rho_t, \nu) \leq \inf_{\gamma \in \Gamma(f_{t\#}\rho_t, \nu)} \int_{S \times Z} j_{t+1}(s^+, z) d\gamma,$$

the backward recursion for the cost-to-go function at time t gives

$$\begin{aligned} J_t(\mu_t, \nu) &= \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \mathbb{E}_{\hat{\xi}_t} [\mathbb{E}_{\rho_t} [g_t(s, a, w)] + J_{t+1}(f_{t\#}\rho_t, \nu)] \\ &\leq \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \mathbb{E}_{\hat{\xi}_t} \left[\underbrace{\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \inf_{\gamma \in \Gamma(f_{t\#}\rho_t, \nu)} \int_{S \times Z} j_{t+1}(s^+, z) d\gamma}_{(*)} \right]. \quad (12) \end{aligned}$$

We now focus on the term denoted by $(*)$. We analyze $(*)$ pointwise for each realization of $\hat{\xi}_t$ before taking expectations. By Lemma B.1, we can “push” the dynamics into the transportation cost, so that

$$(*) = \int_{S \times A \times W} g_t(s, a, w) d\rho_t + \inf_{\gamma' \in \Gamma(\rho_t, \nu)} \int_{S \times A \times W \times Z} j_{t+1}(f_t(s, a, w), z) d\gamma'.$$

Since ρ_t is the first marginal of γ' , Lemma B.2, with $X = S \times A \times W$, $c_1 = g_t$ and $c_2 = j_{t+1} \circ (f_t \times \text{Id}_Z)$, allows to merge the two costs into a single transport objective on $S \times A \times W \times Z$. We therefore have

$$(*) = \inf_{\gamma' \in \Gamma(\rho_t, \nu)} \int_{S \times A \times W \times Z} g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\gamma'.$$

We now use Lemma B.6 with $X = S \times A$, $\rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a}$, and $c(s, a, w, z) = g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)$, to replace the couplings of the full state-action-noise distribution ρ_t with couplings of its $(S \times A)$ -marginal λ_t . This yields the inequality

$$(*) \leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma}.$$

This relaxation removes the dependence on the realized noise from the optimal assignment, thus restricting it to depend only on (s, a) .

We now plug the obtained upper bound for $(*)$ inside the expectation over $\hat{\xi}_t^{s,a}$, as in (12). Since λ_t is fixed and the infimum is now over couplings $\tilde{\gamma} \in \Gamma(\lambda_t, \nu)$ that do not depend on $\hat{\xi}_t^{s,a}$, Lemma B.3 applies, allowing us to interchange expectation and infimum and yielding the upper bound

$$\begin{aligned} \mathbb{E}_{\hat{\xi}_t} [(*)] &\leq \mathbb{E}_{\hat{\xi}_t} \left[\inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma} \right] \\ &\leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \mathbb{E}_{\hat{\xi}_t} \left[\int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma} \right] \\ &= \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} \right] d\tilde{\gamma}, \end{aligned}$$

where the last equality follows from Tonelli's theorem, which applies since the stage cost g_t and the cost-to-go functions j_{t+1} are nonnegative. Informally, the second inequality is precisely when we transition from “choose the assignment after seeing the realized noise” to “choose one assignment against average noise”. We now focus on the inner expectation. In particular, consider a fixed $\tilde{\gamma} \in \Gamma(\lambda_t, \nu)$. We can disintegrate the coupling with respect to its $(S \times A)$ -marginal λ_t as $\tilde{\gamma} = \lambda_t \otimes \kappa_{s,a}$ where $\kappa_{s,a}$ is a (conditional) probability over Z . By non-negativity of all functions, Tonelli's theorem gives

$$\begin{aligned} \int_{S \times A \times Z} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} \right] d\tilde{\gamma} \\ = \int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} d\hat{\xi}_t^{s,a} \right] d\lambda_t. \end{aligned}$$

We can apply Lemma A.9 (treating (s, a) as parameters) to the function

$$\tilde{c}(s, a, w) := \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a}(z).$$

Note, in particular, that Lemma A.9 can be applied since $\kappa_{s,a}$ is fixed and not a function of the noise distribution $\hat{\xi}_t^{s,a}$. Thus, we get

$$\begin{aligned} \int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} d\hat{\xi}_t^{s,a} \right] d\lambda_t \\ = \int_{S \times A} \mathbb{E}_w \left[\int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} \right] d\lambda_t \\ = \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \end{aligned}$$

where the last step follows again from Tonelli's theorem and the definition of $\tilde{\gamma}$. Combined with the previous chain of inequalities, we therefore have

$$\mathbb{E}_{\hat{\xi}_t} [(*)] \leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma}.$$

We can now plug back our bound in (12) to get

$$\begin{aligned} J_t(\mu_t, \nu) &\leq \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \\ &= \inf_{(\text{proj}_{S \times Z}^{S \times A \times Z})_{\#} \tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \\ &=: (*). \end{aligned}$$

Now, we can use Lemma B.4 to disintegrate the optimizer in $(\star)$, which gives

$$(\star) = \inf_{\substack{\tilde{\gamma} \in \Gamma(\mu_t, \nu) \\ \alpha^{s,z} \in \Lambda(A)}} \int_{S \times Z} \int_A \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\alpha^{s,z} d\tilde{\gamma}$$

Since j_t, g_t, f_t are lower semianalytic and $\xi_t(\cdot | s, a)$ is a Borel stochastic kernel, j_t is lower semianalytic by backward induction and so is the integrand. Therefore, we can apply Lemma B.5, which gives

$$(\star) = \inf_{\tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times Z} \inf_{a \in A} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma}.$$

At this point, we recognize that the term in the integral coincides with the Bellman recursion for j_t :

$$j_t(s, z) = \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s, a_t, w_t) + j_{t+1}(f_t(s, a_t, w_t), z)].$$

Hence,

$$J_t(\mu_t, \nu) \leq (\star) = \inf_{\tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\tilde{\gamma}.$$

This concludes the inductive step and the proof of the theorem. $\square$

Two comments on our proof are in order. First, while the proof follows the inductive strategy of the proof of the deterministic separation in [25], there is a key difference: the next population state is random and depends on the realized empirical noise kernel $\hat{\xi}_t$. This requires various additional steps (and preliminary lemmas) in the proof. All these steps trivialize in the absence of noise and, thus, our proof recovers the deterministic separation in [25] when the dynamics are deterministic. Second, our result is an upper bound on the cost-to-go function. Intuitively, this conservatism originates from the following fact: The population state-action distribution can, in principle, choose future couplings *after* observing this realization, while our upper bound predicates that couplings are chosen *before* the noise is realized, averaging over the kernel $\xi_t(\cdot | s, a)$. This is precisely where Lemmas B.3 and B.6 yield inequalities rather than equalities. In the deterministic case, all these steps trivialize and we obtain equality as in [25].

We now prove Part II of the theorem.

Proof. For each timestep t , consider the function

$$\mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma.$$

Let $\varepsilon > 0$. At each time t , we can construct a “separation-based” state-action distribution λ_t^ε as follows: given a population distribution μ_t , choose an $\frac{\varepsilon}{2T}$ -optimal coupling $\gamma_t^\varepsilon \in \Gamma(\mu_t, \nu)$ satisfying

$$\int_{S \times Z} j_t(s, z) d\gamma_t^\varepsilon \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T}, \quad (13)$$

and a universally measurable $\frac{\varepsilon}{2T}$ -optimal action selector $a_t^\varepsilon: S \times Z \rightarrow A$ satisfying, for all (s, z) ,

$$\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] \leq j_t(s, z) + \frac{\varepsilon}{2T}. \quad (14)$$

Formally, we have

$$\lambda_t^\varepsilon := (\text{proj}_S^{S \times Z}, a_t^\varepsilon)_\# \gamma_t^\varepsilon.$$

In words, this state-action distribution assigns agents to targets through γ_t^ε and controls the individual agents through a_t^ε . Clearly, $\lambda_t^\varepsilon \in M(\mu_t)$. In the rest of the proof, we will show that λ_t^ε is ε -optimal, i.e.,

$$J_0^\varepsilon(\mu_0, \nu) := J_0^{\lambda_0^\varepsilon, \dots, \lambda_{T-1}^\varepsilon}(\mu_0, \nu) \leq \mathcal{K}[j_0](\mu_0, \nu) + \varepsilon. \quad (15)$$

Before doing so, we verify well-posedness of the all objects. The $\frac{\varepsilon}{2T}$ -optimal universally measurable selector a_t^ε exists by [29, Proposition 7.50]. For the existence of the $\frac{\varepsilon}{2T}$ -optimal coupling we work as follows. Since j_t is lower semianalytic (by backward induction on (2)), the map $\gamma \mapsto \int_{S \times Z} j_t d\gamma$

is lower semianalytic by [29, Corollary 7.48.1]. Further, the set $\{(\mu, \gamma) : \gamma \in \Gamma(\mu, \nu)\}$ is analytic. Indeed, fixing $\nu \in \mathcal{P}(W)$, define

$$\mathcal{G}_Z := \{(\mu, \gamma) \in \mathcal{P}(S) \times \mathcal{P}(S \times Z) : \gamma \in \Gamma(\mu, \nu)\}.$$

The maps $F_S(\gamma) := (\text{proj}_S^{S \times Z})_{\#} \gamma$ and $F_Z(\gamma) := (\text{proj}_Z^{S \times Z})_{\#} \gamma$ are continuous for the weak topologies, being the pushforward under a continuous map. Hence,

$$\mathcal{G}_Z = \{(\mu, \gamma) : F_S(\gamma) = \mu\} \cap \{(\mu, \gamma) : F_Z(\gamma) = \nu\}.$$

The first set is the preimage of the diagonal in $\mathcal{P}(S) \times \mathcal{P}(S)$ under the continuous map $(\mu, \gamma) \mapsto (\mu, F_S(\gamma))$, and the second is the preimage of the closed singleton $\{\nu\}$ under the continuous map $(\mu, \gamma) \mapsto F_Z(\gamma)$. Since these spaces are Polish, the diagonal and singleton are closed. By continuity F_S and F_Z , the set $\mathcal{G}_\nu$ is closed, hence Borel and in particular analytic. Thus, we can apply [29, Proposition 7.50] to obtain the existence of a universally measurable map

$$\mu \mapsto \gamma_t^\varepsilon(\mu) \in \Gamma(\mu, \nu) \quad \text{with} \quad \int_{S \times Z} j_t d\gamma_t^\varepsilon(\mu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T}.$$

This shows the existence of a universally measurable selector $\gamma_t^\varepsilon(\mu)$. In particular, for fixed μ , $\gamma_t^\varepsilon(\mu)$ is a Borel probability measure and, being the pushforward under a universally measurable (which coincides $\gamma_t^\varepsilon(\mu)$ -a.e. with a Borel map [29, Lemma 7.27]) λ_t^ε is a well-defined Borel probability measure.

We now turn our attention to the desired bound (15). We now show by backward induction the more general bound

$$J_t^\varepsilon(\mu_t, \nu) := J_t^{\lambda_t^\varepsilon, \dots, \lambda_{T-1}^\varepsilon}(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{T-t}{T} \varepsilon, \quad (16)$$

which, evaluated at time $t = 0$, readily implies (15). The base case $t = T$ holds exactly by definition of terminal cost and $j_T = g_T$:

$$J_T^\varepsilon(\mu_T, \nu) = \mathcal{K}[g_T](\mu_T, \nu) = \inf_{\gamma \in \Gamma(\mu_T, \nu)} \int_{S \times Z} g_T(s, z) d\gamma = \mathcal{K}[j_T](\mu_T, \nu).$$

Now fix $t \leq T$ and assume (16) holds at $t + 1$. By the induction hypothesis

$$J_t^\varepsilon(\mu_t, \nu) \leq \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \right] + \frac{T-t-1}{T} \varepsilon. \quad (17)$$

Next, we define

$$\bar{\gamma}_t^\varepsilon := (f_t, \text{Id}_Z)_{\#}(\sigma_{\#} \tilde{\gamma}_t^\varepsilon) \in \mathcal{P}(S \times Z)$$

where

$$\tilde{\gamma}_t^\varepsilon := (\text{Id}_{S \times Z}, a_t^\varepsilon)_{\#} \gamma_t^\varepsilon \otimes \hat{\xi}_t^{s, a_t^\varepsilon(s, z)} \in \mathcal{P}(S \times Z \times A \times W).$$

and we use the permutation consider the permutation

$$\sigma : S \times Z \times A \times W \rightarrow S \times A \times W \times Z, \quad \sigma(s, z, a, w) = (s, a, w, z),$$

to “re-order” the marginals. Informally, $\bar{\gamma}_t^\varepsilon$ is the law of $(f_t(s, a_t^\varepsilon(s, z), w), z)$ where $(s, z) \sim \gamma_t^\varepsilon$ and $w \sim \hat{\xi}_t^{s, a_t^\varepsilon(s, z)}$. We now show $\bar{\gamma}_t^\varepsilon \in \Gamma(\mu_{t+1}, \nu)$. Specifically, the second marginal is ν , since the z -coordinate is unchanged and $(\text{proj}_Z^{S \times Z})_{\#} \gamma_t^\varepsilon = \nu$. The first marginal, instead, is the pushforward of state-action-noise distribution ρ_t^ε associated with λ_t^ε under the dynamics f_t (i.e., the law of $f_t(s, a_t^\varepsilon(s, z), w)$). This is precisely μ_{t+1} , which completes the proof of $\bar{\gamma}_t^\varepsilon \in \Gamma(\mu_{t+1}, \nu)$. By definition of the infimum, we then obtain

$$\mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \leq \int_{S \times Z} j_{t+1}(s, z) d\bar{\gamma}_t^\varepsilon = \int_{S \times Z} \int_W j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z) d\hat{\xi}_t^{s, a_t^\varepsilon(s, z)} d\gamma_t^\varepsilon.$$

Taking expectation over noise realizations, using Fubini-Tonelli’s theorem, and applying Lemma A.9 we get

$$\begin{aligned} & \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \right] \\ & \leq \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times Z} \int_W g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z) d\hat{\xi}_t^{s, a_t^\varepsilon(s, z)} d\gamma_t^\varepsilon \right] \\ & = \int_{S \times Z} \mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] d\gamma_t^\varepsilon. \end{aligned}$$

Then, we use the $\frac{\varepsilon}{2T}$ -optimality of a_t^ε in (14) and of γ_t^ε in (13) to get

$$\begin{aligned} \int_{S \times Z} \underbrace{\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)]}_{\leq j_t(s, z) + \varepsilon/(2T)} d\gamma_t^\varepsilon \\ \leq \int_{S \times Z} j_t(s, z) d\gamma_t^\varepsilon + \frac{\varepsilon}{2T} \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T} + \frac{\varepsilon}{2T}. \end{aligned}$$

We can now combine this bound with (17) and get

$$J_t^\varepsilon(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{T} + \frac{T-t-1}{T}\varepsilon = \mathcal{K}[j_t](\mu_t, \nu) + \frac{T-t}{T}\varepsilon.$$

This concludes the inductive step. The bound at $t = 0$ gives $J_0^\varepsilon(\mu_0, \nu) \leq \mathcal{K}[j_0](\mu_0, \nu) + \varepsilon$. $\square$

Remark C.1 (Existence of optimal couplings under stronger regularity). The proof of Part 2 only requires universally measurable ε -optimal coupling selectors, which exist under the lower semianalytic assumptions used throughout Appendix B. Under the stronger assumption that the cost-to-go functions $j_t : S \times Z \rightarrow \mathbb{R}_{\geq 0}$ are lower semicontinuous and the spaces are compact, the infimum in $\mathcal{K}[j_t](\mu, \nu)$ is attained for each fixed (μ, ν) by the standard existence theorem for optimal couplings [41, Theorem 4.1]. This pointwise existence result is distinct from the measurable-selection argument needed above to construct a Markov policy depending on μ_t .

C.2 Proof of Corollary 3.3

Proof. We construct, for every $\varepsilon > 0$, an admissible separation policy whose cost is within $\frac{(T-t)\varepsilon}{T}$ of $\mathcal{K}[j_t](\mu_t^N, \nu^N)$. Since $\varepsilon > 0$ is arbitrary, and J_t^N is the infimum of the expected cost over all admissible policies, the claim follows. We start by giving an ε -optimal separation policy $\pi_{\text{sep}, t}^\varepsilon$ which attains the upper bound. We will then show that this policy admissible, we directly obtain the desired inequality.

To define $\pi_{\text{sep}, t}^\varepsilon$ we proceed as follows. First, since μ_t^N and ν^N are both empirical measures over N atoms, the optimal transport problem $\mathcal{K}[j_t](\mu_t^N, \nu^N)$ is a finite-dimensional transportation linear program. As such, every coupling between μ_t^N and ν^N can be represented by a doubly-stochastic matrix $\Gamma = (\gamma_{ij})_{i,j=1}^N$ satisfying $\gamma_{ij} \geq 0$, $\sum_{i=1}^N \gamma_{ij} = \sum_{j=1}^N \gamma_{ij} = \frac{1}{N}$. By the Birkhoff-von Neumann theorem [42], every doubly stochastic matrix is a convex combination of permutation matrices; since the objective is linear, the minimum is attained by a scaled permutation matrix. Thus,

$$\mathcal{K}[j_t](\mu_t^N, \nu^N) = \frac{1}{N} \min_{\sigma_t \in S_N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma_t(i)}),$$

where S_N denotes the group of permutations of $\{1, \dots, N\}$. Let σ_t^* attain this minimum and write $z_t^{*,i} := z^{\sigma_t^{*,i}}$ for the target assigned to agent i . Given that agents are now assigned to targets, we can now consider the single-agent optimal control problem from s_t^i to the corresponding target $z_t^{*,i}$. Specifically, for $\varepsilon > 0$, there exists a universally measurable $\frac{\varepsilon}{T}$ -optimal action selector $a_t^\varepsilon : S \times Z \rightarrow A$, satisfying, for all $(s, z) \in S \times Z$:

$$\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] \leq j_t(s, z) + \frac{\varepsilon}{T}.$$

At this point, we define the separation policy $\pi_{\text{sep}, t}^\varepsilon$ at time t . An agent at s_t^i executes the action $a_t^\varepsilon(s_t^i, z_t^{*,i})$. Informally, the optimal transport problem indicates the target, whereas the single-agent optimal control problem determines the optimal action to reach that target. Clearly, this policy is deterministic and Markovian, as it solely depends on $\mathbf{s}_t = (s_t^1, \dots, s_t^N)$. Measurability follows from the measurability arguments in the proof of Theorem 3.1.

At this point, it suffices to verify that this policy yields the bound on the cost. In particular, we show that the expected cost-to-go function under this policy, denoted by $J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N)$, is upper bounded as follows:

$$J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N) + \frac{T-t}{T}\varepsilon,$$

which gives the desired upper bound. We proceed by induction. The equality at time $t = T$ is trivial, as the terminal cost does not depend on the policy but only on the terminal state. Suppose then the statement holds at some time $t + 1$. Then,

$$\begin{aligned}
J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) &\leq \mathbb{E}_{w_t^i} \left[\frac{1}{N} \sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) + \mathcal{K}[j_{t+1}](\mu_{t+1}^N, \nu^N) \right] + \frac{T-t-1}{T} \varepsilon \\
&= \frac{1}{N} \mathbb{E}_{w_t^i} \left[\sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) + \min_{\sigma_{t+1} \in S_N} \sum_{i=1}^N j_{t+1}(f_t(s_t^i, a_t^i, w_t^i), z^{\sigma_{t+1}(i)}) \right] \\
&\quad + \frac{T-t-1}{T} \varepsilon \\
&\leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{w_t^i} \left[g_t(s_t^i, a_t^i, w_t^i) + j_{t+1}(f_t(s_t^i, a_t^i, w_t^i), z^{\sigma_t^*(i)}) \right] + \frac{T-t-1}{T} \varepsilon \\
&\leq \frac{1}{N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma_t^*(i)}) + \frac{\varepsilon}{T} + \frac{T-t-1}{T} \varepsilon \\
&= \frac{1}{N} \min_{\sigma \in S_N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma(i)}) + \frac{T-t}{T} \varepsilon,
\end{aligned}$$

where the first inequality is precisely the inductive assumption, the second inequality follows since σ_t^* is a potentially suboptimal permutation, last inequality follows from the use of the policy $\pi_{\text{sep},t}^\varepsilon$, and the last equality the definition of σ^* . This gives

$$J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N) + \frac{T-t}{T} \varepsilon,$$

which completes the induction.

Since $\varepsilon > 0$ is arbitrary, and J_t^N is the infimum of the expected cost over all admissible policies, letting $\varepsilon \downarrow 0$ gives

$$J_t^N(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N).$$

□

D Additional numerical results

This appendix provides complementary experimental results and full environment and hyperparameter specifications for the experiments in Section 5. Section D.1 covers the grid-world setup and additional metrics. Section D.2 details the scalability experiment. Section D.3 describes the Manhattan road network environment.

D.1 Additional details for the grid-world experiments in Section 5.1

Environment. The environment consists of an 11×30 grid with horizon $T = 100$ and $N = 16$ agents. At the beginning of each episode, agents are initialized uniformly at random in the first column, while the N targets are sampled uniformly in the last ten columns. The action space is $\mathcal{A} = \{\text{up}, \text{down}, \text{left}, \text{right}, \text{stay}\}$. At each timestep, the intended action is executed with probability $1 - p$ and replaced by a uniformly sampled action with probability p . We evaluate noise probabilities $p \in \{0, 0.1, 0.25\}$ under individual, local, and global noise. Individual noise samples this perturbation independently for each agent, local noise shares a perturbation among agents with the same state-action pair, and global noise shares one perturbation across the whole population. Their formal definitions are provided in Appendix A. After the transition, positions are clipped to remain inside the grid. In both MARL and the target-conditioned single-agent control problem in SALT, we introduce an induced bias to facilitate learning. For the MARL baselines, an agent becomes inactive when it first reaches an unoccupied target. For SALT, learning is target-conditioned and single-agent: the agent is associated with one assigned target z , and becomes inactive when it reaches that assigned target (note, in particular, that this is not coupling agents, but rather facilitates the learning of the single-agent target-conditioned problem). Absorbed agents incur zero further stage cost, ignore future actions and perturbations, and remain fixed for the rest of the episode.

Reward structure. Each active agent incurs a stage cost of 1 at every timestep. At the horizon, an additional optimal transport cost $K[c](\mu_T, \nu)$ is applied, where the cost function c is the Manhattan distance between grid cells. The combined objective therefore penalizes both travel time and terminal coverage quality.

Baselines. We compare against MAPPO and IPPO, VDN, and QMIX. All baselines use the same episode budget and random seeds. All neural policies use two hidden layers with 64 units and tanh activations. The separation-based neural methods are trained as goal-conditioned single-agent controllers. For SALT-PPO, SALT-A2C, and SALT-DDQN, the policy input is

$$\left(\frac{r}{H_g}, \frac{c}{W_g}, \frac{r_z}{H_g}, \frac{c_z}{W_g}, \frac{t}{T-1}, \frac{\|s - z\|_1}{H_g + W_g} \right),$$

where (r, c) is the current agent position, (r_z, c_z) is the currently assigned target, t is the timestep, h and w are the grid height and width, and $H_g = h - 1$, $W_g = w - 1$ are the maximum row and column indices. Thus, the separation policy never observes the full population during action selection; coordination of the agents enters only through the optimal transport assignment that chooses the target z .

MAPPO, IPPO, QMIX, and VDN receive target-relative local observations. For each agent i , the local observation has dimension $5 + 3M$, where $M = N$ is the number of targets:

$$\left(\frac{r_i}{H_g}, \frac{c_i}{W_g}, \frac{i}{N-1}, \frac{t}{T-1}, a_i, \frac{\Delta r_{i1}}{H_g}, \frac{\Delta c_{i1}}{W_g}, b_1, \dots, \frac{\Delta r_{iM}}{H_g}, \frac{\Delta c_{iM}}{W_g}, b_M \right).$$

Here a_i is an active-agent indicator, $\Delta r_{ij} = r_{z_j} - r_i$ and $\Delta c_{ij} = c_{z_j} - c_i$ are the relative coordinates of target j from agent i , and b_j indicates whether target j remains unoccupied. The target set is therefore visible to each baseline agent as an ordered list of relative target offsets and target-availability flags. MAPPO additionally uses a centralized critic state of dimension $3N + 1 + 3M$, containing all normalized agent positions, all agent active flags, the timestep, all normalized target positions, and all target-availability flags. QMIX uses the same global state inside its monotone mixing network with mixer embedding dimension 32. IPPO and VDN do not use a separate global critic state; IPPO uses a local critic, and VDN uses additive value decomposition. All neural methods are optimized with Adam.

Hyperparameters We summarize in Tables 2, 3, and 4 the used hyperparameters.

Table 2: Shared grid-world training budget and evaluation settings for Section 5.1.

Quantity	Value
Grid / horizon	$11 \times 30, T = 100$
Agents / targets	$N = 16$
Training seed	0
Evaluation seeds	20 paired seeds
Noise probabilities	0, 0.1, 0.25
Noise modes	individual, local, global
Training episodes	320000
Environment interactions	$320000 \times 16 \times 100 = 5.12e8$
Batch episodes	64
Update batches for neural methods	5000
Step cost	1
Evaluation target region	columns 20–29
Evaluation rematching for SALT	every timestep

Additional performance metrics Finally, we here complement the average travel time results reported in Figure 2. Since the task requires both successful navigation and coordination among agents, we evaluate the algorithms using several complementary metrics. The average travel time of the population in Figure 2 measured *how quickly* the agents covered the target distribution. Here, we focus on understanding *how well* and *how successfully* they cover it. Figure 4 gives a measure of

Table 3: Hyperparameters used for SALT in Section 5.1.

Hyperparameter	SALT-Q	SALT-DDQN	SALT-PPO	SALT-A2C
Discount γ	0.98	0.98	0.98	0.98
Learning rate	0.25 $\rightarrow$ 0.02	$3 \cdot 10^{-4}$	actor $3 \cdot 10^{-4}$ critic 10^{-3}	actor $3 \cdot 10^{-4}$ critic 10^{-3}
Hidden size	—	64	64	64
GAE λ	—	—	0.95	0.95
PPO clip / epochs	—	—	0.2 / 4	—
Entropy coefficient	—	—	0.01	0.01
Max grad norm	—	0.5	0.5	0.5
Exploration α	0.5 $\rightarrow$ 0.05	0.5 $\rightarrow$ 0.05	0.5 $\rightarrow$ 0.05	0.5 $\rightarrow$ 0.05
Exploration decay (interactions)	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$
Replay capacity	—	500000	—	—
Replay warmup / minibatch	—	5000 / 256	—	—
Learner steps per batch	—	64	—	—
Target update period	—	500 learner steps	—	—

Table 4: Hyperparameters used for MARL baselines in Section D.1.

Hyperparameter	MAPPO	IPPO	QMIX	VDN
Discount γ	0.98	0.98	0.98	0.98
Learning rate	actor $3 \cdot 10^{-4}$ critic 10^{-3}	actor $3 \cdot 10^{-4}$ critic 10^{-3}	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$
Hidden size	64	64	64	64
GAE λ	0.95	0.95	—	—
PPO clip / epochs	0.2 / 4	0.2 / 4	—	—
Entropy coefficient	0.01 $\rightarrow$ 0.001	0.01 $\rightarrow$ 0.001	—	—
Max grad norm	0.5	0.5	0.5	0.5
Actor minibatch	4096	4096	—	—
Critic minibatch	1024	4096	—	—
Replay capacity	—	—	100000	100000
Minimum replay size	—	—	2048	2048
Replay batch size	—	—	512	512
Updates per batch	—	—	1	1
Target update period	—	—	50 batches	50 batches
Exploration α	—	—	0.5 $\rightarrow$ 0.05	0.5 $\rightarrow$ 0.05
Exploration decay	—	—	160000 episodes	160000 episodes
QMIX mixer embedding	—	—	32	—

coverage quality by reporting the empirical optimal transport cost between the final agent positions and the set of targets. This metric measures the minimum average Manhattan distance required to match each agent to a target. Let

$$\mu_T^N = \frac{1}{N} \sum_{i=1}^N \delta_{s_T^i}, \quad \nu^N = \frac{1}{N} \sum_{j=1}^N \delta_{z^j}.$$

We report the empirical terminal optimal transport cost

$$\mathcal{K}[d](\mu_T^N, \nu^N) = \min_{\sigma \in S_N} \frac{1}{N} \sum_{i=1}^N d(s_T^i, z^{\sigma(i)}),$$

where S_N is the set of permutations of $\{1, \dots, N\}$, s_T^i denotes the final position of agent i , z^j denotes the target locations, $d(\cdot, \cdot)$ is the Manhattan distance, and σ is the optimal assignment between agents and targets. Hence, a value of zero indicates that every target is perfectly occupied by an agent. This allows us to distinguish between different agents failures: populations that almost reached all targets should be valued more positively than populations that just moved randomly.

Finally, Figure 5 focuses on the agents' ability to successfully navigate the goals within the episode horizon. Specifically, the reach rate quantifies the fraction of agents that reach a target during the

	Individual noise			Local noise			Global noise		
SALT-Q	0.65	0.65	0.50	0.65	0.70	0.00	0.65	0.50	0.50
SALT-DDQN	0.55	0.15	0.15	0.55	0.00	0.60	0.55	1.90	1.90
SALT-PPO	0.95	0.55	1.70	0.95	0.15	1.40	0.95	1.35	2.05
SALT-A2C	0.65	0.80	1.20	0.65	0.00	0.20	0.65	0.50	0.15
MAPPO	12.75	12.10	14.90	12.75	12.95	14.65	12.75	13.30	14.85
IPPO	10.35	14.70	20.05	10.35	9.65	12.40	10.35	14.00	14.05
QMIX	20.30	23.55	24.75	20.30	21.60	30.30	20.30	22.45	21.95
VDN	20.70	31.35	25.55	20.70	35.90	28.25	20.70	22.05	22.75
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 4: Optimal transport cost between the final agent positions and the set of targets against environment noise for our separation principle.

	Individual noise			Local noise			Global noise		
SALT-Q	100	99.3	100	100	99.3	100	100	100	99.3
SALT-DDQN	100	100	100	100	100	100	100	100	100
SALT-PPO	100	100	100	100	100	100	100	100	100
SALT-A2C	100	100	100	100	100	100	100	100	100
MAPPO	77.1	76.4	70.7	77.1	78.6	73.6	77.1	75.7	75.7
IPPO	85.7	78.6	71.4	85.7	81.4	75	85.7	80	79.3
QMIX	56.4	52.9	58.6	56.4	48.6	42.1	56.4	56.4	56.4
VDN	55	33.6	52.1	55	30.7	38.6	55	50	51.4
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 5: Reach rate against environment noise. Entries are means and darker cells indicate better relative performance within this metric.

episode. Figure 6 summarizes the three evaluation metrics used in Figures 2, 5 and 4: average travel time, terminal optimal transport cost, and reach rate. Each value is averaged over the same paired evaluation seeds for a fixed algorithm, noise model, and noise probability p . The metrics are reported on their native scales: higher is better for reach rate, while lower is better for terminal optimal transport cost and average travel time. Across all three metrics, the SALT variants achieve higher reach rates and lower terminal optimal transport costs and travel times than the MARL baselines.

D.2 Additional details for the scalability experiment

To study sample complexity, we consider square grids. All N agents are initialized in the left-bottom corner, and targets are uniformly sampled in the whole grid. To isolate population-size scaling from task difficulty, we scale grid dimensions proportionally with N so that target density remains approximately constant (see Table 5). All experiments are conducted under the individual-noise

Table 5: Population sizes and grid sizes used in the scalability experiment.

Agents N	2	4	6	8	16	32	64	128
Grid size	10^2	14^2	17^2	20^2	28^2	40^2	57^2	80^2
Horizon	20	28	34	40	56	80	114	160
Planned batches	781250	279018	153186	97656	34877	12207	4283	1526

model with noise probability $p = 0.1$. We instantiate SALT with PPO as the single-agent learner and compare it against MAPPO. The hyperparameters for both methods follow Table 3 and Table 4 above.

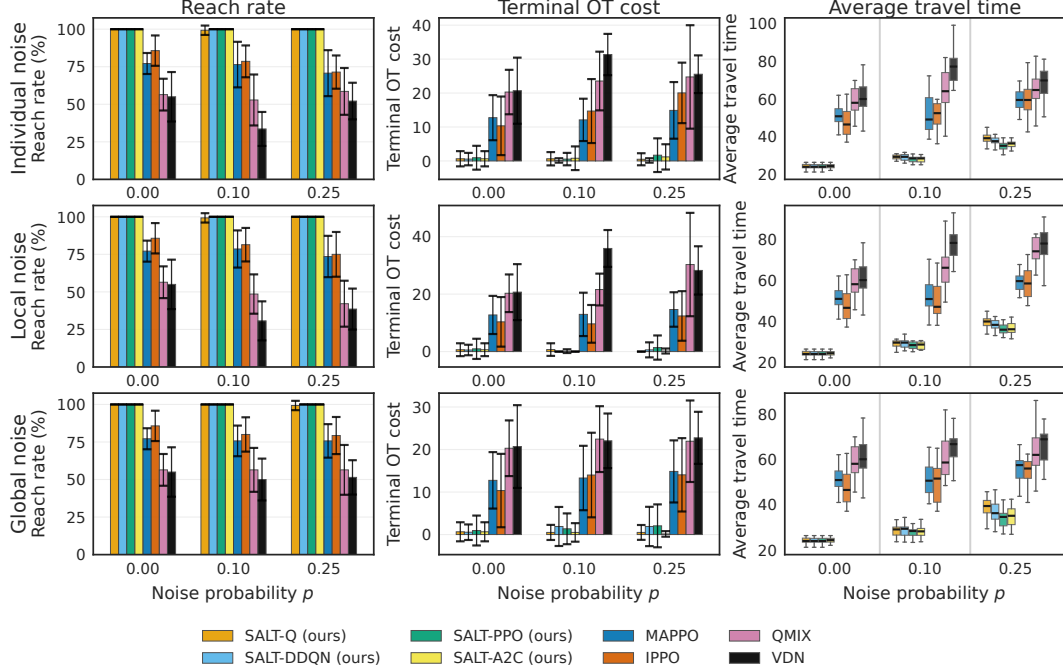


Figure 6: Combined reach rate, terminal optimal transport cost and average travel time.

The batch size is 64 episodes; the number of batches is chosen separately for each population size so that the interaction budget is approximately constant and equal to 2×10^9 .

D.3 Towards a realistic, time-varying urban logistics task

Manhattan environment and dynamics. The Manhattan environment is a directed road graph extracted from OpenStreetMap [36] for southern Manhattan. OpenStreetMap data are made available under the Open Data Commons Open Database License (ODbL). In the current training run, the processed graph contains $|V| = 2088$ nodes. Each node corresponds to an intersection and each directed edge to an admissible road segment. Nominal edge travel times are obtained from road length and road-type-dependent speeds.

Traffic signals are modeled at the edge level. Each edge has a traffic-light cycle of 90 seconds. Green-phase durations depend on road type: primary roads are green for 50% of the cycle, secondary and residential roads for 30%, and motorway, trunk, tertiary, and unclassified roads are always green. Each signal receives a randomized phase offset. When an agent traverses edge e at time t , the realized transition time is the travel time plus the signal waiting time induced by the edge’s phase.

We additionally inject stochastic congestion noise on selected road types. For a traversed edge e , the travel time is multiplied by

$$1 + u_t, \quad u_t \sim \text{Unif}(0, \eta_e),$$

where $\eta_e = 0.1$ for primary roads and $\eta_e = 0.15$ for secondary roads and u_t is independently drawn across edges.

Time discretization and Q-table. The Manhattan simulator evolves in event time: one decision step corresponds to traversing one outgoing road segment, not to advancing by a fixed one-second tick. The tabular Q-learner uses a discrete time coordinate with resolution

$$\Delta t = 1 \text{ second.}$$

Since all traffic lights use a 90 second cycle, the Q-table contains $T_{\text{disc}} = 90/\Delta t = 90$ time slices. The stored time coordinate is the traffic phase, not the unbounded absolute clock time. Thus, for physical time t and cycle period $P = 90$ seconds, the table index is $\tau(t) = (\text{round}(t/\Delta t)) \bmod T_{\text{disc}}$.

The Q-table is target-conditioned and indexed by current node v , pickup node z , traffic phase τ , and outgoing action a :

$$Q(v, z, \tau, a).$$

In the Manhattan graph this gives shape

$$|V| \times |V| \times T_{\text{disc}} \times \max_v |\text{out}(v)| = 2088 \times 2088 \times 90 \times 4,$$

where $|\text{out}(v)|$ is the maximum outgoing degree of the graph. Invalid padded actions are masked during action selection and Bellman backups. The table is stored in `bfloat16` to reduce memory usage.

For consistency with the tabular state space, edge travel times are rounded to the same grid before training. If $T_{v,a}$ is the continuous travel time of edge (v, a) , the learner uses $\hat{T}_{v,a} = \Delta t \text{round}(T_{v,a}/\Delta t)$. After traversing the edge, the simulator computes the traffic-light waiting time from the resulting arrival phase. For edge period $P_{v,a}$, offset $o_{v,a}$, and green duration $g_{v,a}$,

$$\phi = (t + \hat{T}_{v,a} + o_{v,a}) \bmod P_{v,a},$$

and

$$W_{v,a}(t) = \begin{cases} 0, & \phi < g_{v,a}, \\ P_{v,a} - \phi, & \phi \geq g_{v,a}. \end{cases}$$

The next physical time is

$$t' = t + \hat{T}_{v,a} + W_{v,a}(t),$$

and the next stored time is rounded back to the discrete grid and reduced modulo the traffic-light period:

$$\tilde{t}' = (\Delta t \text{round}(t'/\Delta t)) \bmod P_{v,a}.$$

This $\tilde{t}'$ is the time component used for the next Q-table lookup. The reward uses the realized delay $\hat{T}_{v,a} + W_{v,a}(t)$, scaled in minutes, plus the pickup bonus when the target is reached.

Task setup. Starts are sampled from three fixed depot nodes. These depots are selected once before training using betweenness centrality, with a preference for nodes in different taxi zones when zone metadata is available. Pickup locations are sampled uniformly from all graph nodes.

Before routing, pickups are assigned to agents by solving a linear assignment problem with the Hungarian algorithm. Since starts and pickups have equal mass, this is the empirical optimal-transport problem. The assignment cost is the negative current target-conditioned Q-value estimate, used as a proxy for the single-agent cost-to-go function j_t .

Learning algorithm. The Manhattan experiment uses tabular Q-learning. The Q-learning update uses learning rate 0.1 and discount factor $\gamma = 1.0$. Exploration is ε -greedy, with

$$\varepsilon_{\text{start}} = 1.0, \quad \varepsilon_{\text{end}} = 0.01.$$

The decay is linear over 90% of the total training steps. The training budget is 5,000,000 episodes.

The Q-table is initialized from offline shortest-path travel-time estimates (hence blind to noise and time-varying aspects) before learning. In the current configuration, initialization is applied to the initial time slice only, rather than all 90 time slices, to avoid the large memory and compilation cost of full time-slice initialization.

Reward. At each step, an active agent receives the negative realized delay in minutes plus a pickup bonus upon reaching the target:

$$r_t = -\frac{\text{travel}_t + \text{wait}_t}{60} + 5 \cdot \mathbf{1}\{\text{pickup reached}\}.$$

Completed agents enter an absorbing state and receive zero reward thereafter.

Evaluation and logging. Training metrics are logged every 1,000 episodes. Evaluation is performed every 10,000 episodes. Final evaluation samples 500 sets of starts and pickups and evaluates all set combinations using Hungarian matching. Experiments are run with JAX [43].

D.4 Compute resources

All reported experiments were run on a single-GPU workstation. Unless otherwise stated, each run used one NVIDIA GeForce RTX 4090 GPU with 24 GB of GPU memory (24564 MiB as reported by the system) and 450 W maximum board power. The machine used 20 physical CPU cores and 28 logical CPUs. The experiments were run on Linux 6.8.0-90 with CPython 3.10.12.

For the Manhattan road-network experiments, a representative end-to-end run used four parallel workers and ran for 46.26 GPU-hours. All grid-world experiments were run on Apple M3 Max MacBook Pro, 14 CPU cores. The average training job took 40.3 minutes, with an average of 13.2 minutes for our first experiment (performance) and 2.70 hours for the second experiment (scalability). A full experimental condition, including all algorithms trained for that setting, took 3.47 hours on average (1.76 hours for performance and 5.40 hours for scalability).

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